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INTRODUCTION TO FINANCIAL MODELS

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Portfolio Optimization Using the Markowitz Model

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Portfolio Optimization Using the Markowitz Model

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Abstract

This study applies the Markowitz mean–variance framework to construct optimal equity portfolios using stocks listed on the Cambodia Securities Exchange (CSX). Daily price data are used to estimate expected returns and the covariance matrix of asset returns. Portfolio optimization is performed under both short-selling and long-only constraints using analytical and numerical methods. The efficient frontier and Global Minimum Variance (GMV) portfolio are examined, and portfolio performance is evaluated through out-of-sample backtesting. Sensitivity analyses assess the effects of correlation, volatility, and expected return assumptions on portfolio stability.Results show that GMV portfolios achieve significant risk reduction relative to equal-weighted benchmarks, while short selling provides limited practical benefits in the CSX context. Overall, the findings support the use of robust, covariance-based portfolio strategies for investors in emerging markets.

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1 Introduction

Imagine placing all eggs in a single basket. If the basket falls, all the eggs break. However, if the eggs are placed in several baskets, a fall may damage only part of them. This simple idea captures the essence of diversification in investment. In financial markets, investing all capital in a single asset exposes an investor to substantial risk, as poor performance or adverse market conditions affecting that asset can lead to significant losses. Diversification, by spreading investments across multiple assets, helps reduce this risk.

In finance, the return on a risky investment is uncertain and depends on future market conditions. As a result, risk must be quantified in a meaningful way. Modern portfolio theory, introduced by **Harry Markowitz**, measures risk using the variance (or standard deviation) of returns, which captures both the dispersion of possible outcomes and their probabilities. A higher variance indicates greater uncertainty and therefore higher risk, while a lower variance reflects more stable returns.

Markowitz's portfolio theory shows that the risk of a portfolio depends not only on the individual risks of the assets it contains, but also on the correlations between their returns. By combining assets with different risk characteristics and imperfect correlations, it is possible to construct portfolios that achieve lower risk without sacrificing expected return. This leads to the concept of the efficient frontier, which represents the set of portfolios offering the highest expected return for a given level of risk.

The purpose of this project is to apply the Markowitz mean–variance optimization framework to stocks listed on the Cambodia Securities Exchange (CSX). Using historical price data, expected returns and the covariance matrix of asset returns are estimated to construct optimal portfolios under different constraints, including cases with and without short selling. Through this analysis, the project illustrates the practical benefits of diversification and demonstrates how modern portfolio theory can be applied in an emerging market context such as Cambodia.

2 Research Objectives

The main objective of this study is to apply modern portfolio theory to stocks listed on the Cambodia Securities Exchange (CSX) in order to determine optimal portfolio allocations under different risk–return considerations. Specifically, this project aims to:

1. Compute the sample mean vector and covariance matrix of asset returns.
2. Solve the Markowitz optimization problem with and without short-selling constraints.
3. Plot the efficient frontier showing the minimum variance and maximum return portfolios.
4. Analyze how correlations between assets influence the shape of the efficient frontier.

3 The Overview of Markowitz Mean–Variance Portfolio Framework

In this framework, **risk** is measured by the **variability of returns**, while **return** is measured by the **expected value of those returns**. A portfolio is therefore evaluated by two quantities: its expected return and its variance. Investors prefer portfolios that offer higher expected returns for a given level of risk, or lower risk for a given level of expected return.

A key insight of the Markowitz model is that portfolio risk depends not only on the risk of individual assets, but also on how their returns move together. By combining assets that do not move in the same direction at the same time, investors can reduce total portfolio risk through diversification.

3.1 Key Inputs of the Model

To apply the mean–variance framework in practice, three main inputs are required. All of these inputs are estimated from historical price data.

Asset Returns. Let $P_t^{(i)}$ denote the closing price of asset i at time t. Asset returns are computed using logarithmic returns,

$$r_t^{(i)} = \ln\left(\frac{P_t^{(i)}}{P_{t-1}^{(i)}}\right).$$

Logarithmic returns are widely used in economics and finance because they are time-additive and allow price changes over different periods to be compared consistently. They also provide a stable basis for measuring risk using variance.

Expected Returns. The expected return of each asset is estimated as the historical average of its logarithmic returns. These expected returns are collected into a vector

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^\top,$$

which represents the average performance of each asset over the sample period. This vector captures the return component of the risk–return trade-off.

Covariance Matrix. Portfolio risk depends on how asset returns move relative to one another. This relationship is captured by the covariance matrix,

$$\Sigma = \left[\sigma_{ij}\right], \qquad \sigma_{ij} = \mathrm{Cov}(r^{(i)}, r^{(j)}).$$

The diagonal elements of Σ measure the variance of individual assets, while the off-diagonal elements measure how pairs of assets move together. Using this matrix, portfolio variance is given by

$$\sigma_V^2 = \mathbf{w}^\top \Sigma \mathbf{w},$$

where $\mathbf{w}$ denotes the vector of portfolio weights.

4 Data Acquisition & Pre-processing

4.1 Dataset Description

The dataset used in this study is obtained from the Cambodia Securities Exchange (CSX). It consists of historical daily closing prices of **10** selected companies listed on the exchange, covering the period from January 2021 to December 2025. These companies operate in key sectors of the Cambodian economy, including banking, infrastructure, utilities, manufacturing, and education. The selected stocks provide a diversified representation of the CSX market, making them suitable for portfolio optimization analysis under the Markowitz framework and subsequent CAPM-based evaluation.

Table 1: CSX-Listed Companies Used in the Study

Acronym	Full Name	Description
ABC	ACLEDA Bank Plc	A leading commercial bank in Cambodia providing retail and corporate banking services.
GTI	Grand Twins International (Cambodia) Plc	A garment manufacturing company focused on export-oriented clothing production.
PAS	Sihanoukville Autonomous Port	The main international seaport handling cargo and trade logistics.
PEPC	Pestech (Cambodia) Plc	An engineering firm specializing in power and infrastructure projects.
PPAP	Phnom Penh Autonomous Port Plc	An inland port operator supporting river-based cargo transport.
PPSP	Phnom Penh SEZ Plc	An operator of a special economic zone supporting industrial development.
PWSA	Phnom Penh Water Supply Authority	A public utility responsible for urban water treatment and distribution.
CGSM	Cambodia Gas Supply Management Co., Ltd.	A firm engaged in LPG import, storage, and distribution.
MJQE	MJQ Education Plc	An education provider operating international schools in Cambodia.

4.1.1 Handling Time-Series Discrepancies

The selected CSX-listed stocks do not share a common trading history. While some assets have data available from 2021, others, such as MJQE and CGSM, began trading only in mid–2023. Using unequal sample lengths would lead to inconsistent estimates of returns and covariances.

To ensure comparability across assets, all price series were aligned to a common observation window from June 28, 2023 to December 17, 2025. Restricting the analysis to this shared period guarantees that expected returns and the covariance matrix are computed from the same market conditions.

4.1.2 Data Cleaning and Return Construction

After aligning all price series to a common time window, the next step is to construct a clean and consistent return dataset suitable for portfolio optimization.

Daily asset returns were computed using logarithmic returns, defined as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right),$$

where P_t denotes the closing price on day t. This transformation converts price levels into stationary return series and ensures compatibility with variance-based risk modeling. (Capiński & Zastawniak, 2003).

The calculation of logarithmic returns naturally introduces missing values in the first observation of each series, as no prior price is available. These initial missing values were removed to obtain a complete return matrix.

Table 2: Logarithmic Returns Before and After Data Cleaning (Excerpt)

Date	**ABC**	**GTI**	**PAS**	**PEPC**	**PPAP**	**MJQE**
Before cleaning						
2023-06-28	NaN	NaN	NaN	NaN	NaN	NaN
2023-06-29	0.0000	0.0065	0.0062	0.0000	-0.0028	-0.0323
2023-06-30	0.0000	-0.0032	0.0046	-0.0137	0.0000	0.0000
After cleaning						
2023-06-29	0.0000	0.0065	0.0062	0.0000	-0.0028	-0.0323
2023-06-30	0.0000	-0.0032	0.0046	-0.0137	0.0000	0.0000
2023-07-03	-0.0020	0.0161	-0.0015	0.0069	-0.0070	-0.0094

The table shows that missing values occur only in the first row of the return matrix due to the logarithmic transformation. After removing this row, the dataset becomes complete, containing 607 daily observations per asset with no remaining missing values. This cleaned return matrix is then used for all subsequent statistical analysis and portfolio optimization.

Finally, descriptive statistics were computed for the cleaned return series to verify their numerical properties and assess basic distributional features such as mean, volatility, and extreme values.

Table 3: Descriptive Statistics of Daily Logarithmic Returns

Asset	**Mean**	**Std. Dev.**	**Min**	**Max**
ABC	-0.0006	0.0080	-0.0964	0.0398
GTI	0.0014	0.0175	-0.1054	0.0939
PAS	0.0000	0.0055	-0.0364	0.0260
PEPC	-0.0001	0.0207	-0.0705	0.0953
PPAP	-0.0000	0.0094	-0.0325	0.0949
PPSP	-0.0000	0.0089	-0.0357	0.0940
PWSA	-0.0003	0.0054	-0.0502	0.0282
CGSM	0.0001	0.0120	-0.1044	0.0953
MJQE	-0.0002	0.0074	-0.0654	0.0949

Table 3 summarizes the descriptive statistics of daily logarithmic returns for the selected CSX-listed stocks. Mean returns are close to zero for most assets, indicating modest average price changes, a common feature of emerging equity markets.

Volatility differs across assets. Stocks such as PEPC and GTI exhibit higher standard deviations, reflecting greater price fluctuations, while PAS and PWSA show lower volatility and more stable return behavior. These differences are important for portfolio construction, as lower-volatility assets typically receive higher weights in risk-minimizing portfolios.

The observed minimum and maximum returns indicate that occasional sharp price movements occur despite generally low daily volatility, highlighting the role of liquidity constraints and reinforcing the importance of diversification.

4.1.3 Outliers Treatment

The Markowitz mean–variance framework relies on the assumption that asset returns are reasonably well approximated by a normal distribution, such that risk can be summarized using the mean and variance alone. In empirical applications, however, return distributions often exhibit extreme observations caused by thin trading, delayed price adjustments, or market microstructure effects. These outliers distort estimated moments and may lead to unstable portfolio weights.

To assess the validity of the normality assumption, return distributions are first examined prior to any treatment.

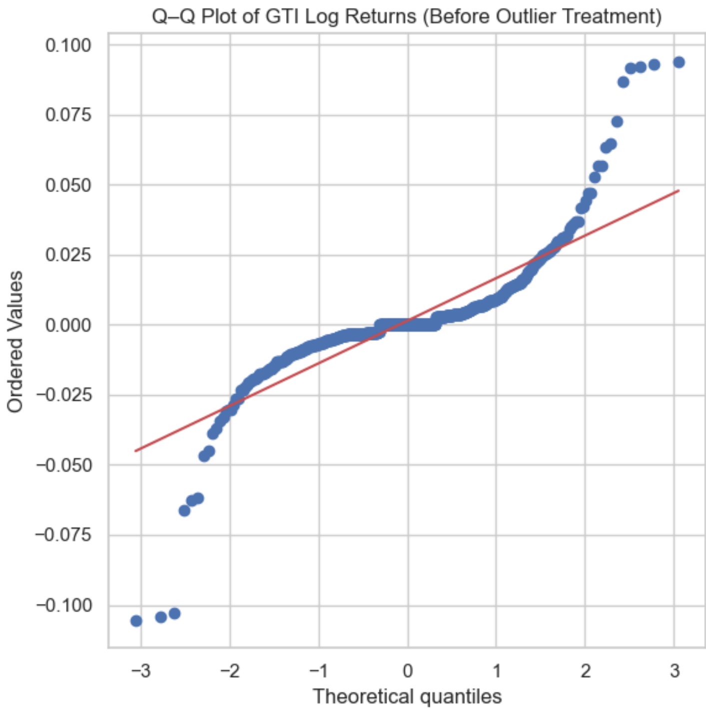

Figure 1: Q–Q plot of selected asset log returns before outlier treatment. Deviations from the 45-degree line indicate heavy tails and departures from normality.

The presence of heavy tails motivates the application of an outlier containment procedure. Rather than removing observations, this study applies *winsorization*. **Winsorization** works by placing a reasonable limit on how extreme a return is allowed to be. For each asset, the sample mean and standard deviation of daily log returns are first computed. A lower and an upper bound are then defined as the **mean minus three standard deviations** and the **mean plus three standard deviations**, respectively. Any return that falls outside this interval is not removed from the dataset; instead, it is replaced by the **nearest boundary value**.

The choice of three standard deviations follows a common rule in statistics: under approximate normality, more than 99% of observations should lie within this range.

Values beyond this threshold are therefore likely to represent extreme or irregular price movements rather than typical market behavior. By capping these extremes, winsorization reduces the influence of abnormal spikes on the estimated variance and covariance, while preserving the number of observations and the overall time-series structure of returns.

Table 4: Summary Statistics of Daily Logarithmic Returns After Winsorization

Statistic	**ABC**	**GTI**	**PAS**	**PEPC**	**PPAP**	**PPSP**	**PWSA**	**CGSM**	**MJQE**
Count	607	607	607	607	607	607	607	607	607
Mean	-0.0004	0.0013	0.0001	-0.0003	-0.0003	-0.0003	-0.0002	-0.0001	-0.0002
Std. Dev.	0.0062	0.0144	0.0050	0.0196	0.0079	0.0071	0.0046	0.0078	0.0050
Min	-0.0246	-0.0513	-0.0166	-0.0621	-0.0283	-0.0269	-0.0163	-0.0359	-0.0223
25%	-0.0027	-0.0036	-0.0017	-0.0083	-0.0029	-0.0046	-0.0028	-0.0040	-0.0046
Median	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
75%	0.0021	0.0046	0.0018	0.0076	0.0027	0.0045	0.0028	0.0041	0.0000
Max	0.0235	0.0540	0.0166	0.0619	0.0283	0.0268	0.0158	0.0362	0.0219

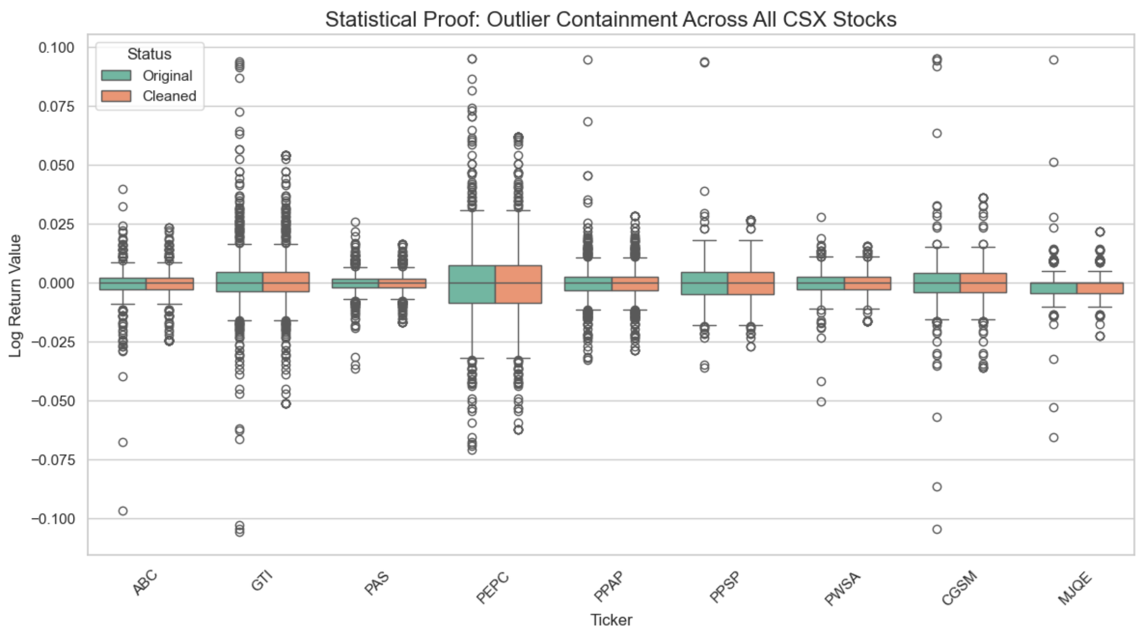

Figure 2: Comparison of return distributions before and after winsorization. The cleaned series exhibit tighter dispersion, indicating effective containment of extreme outliers across all CSX-listed stocks.

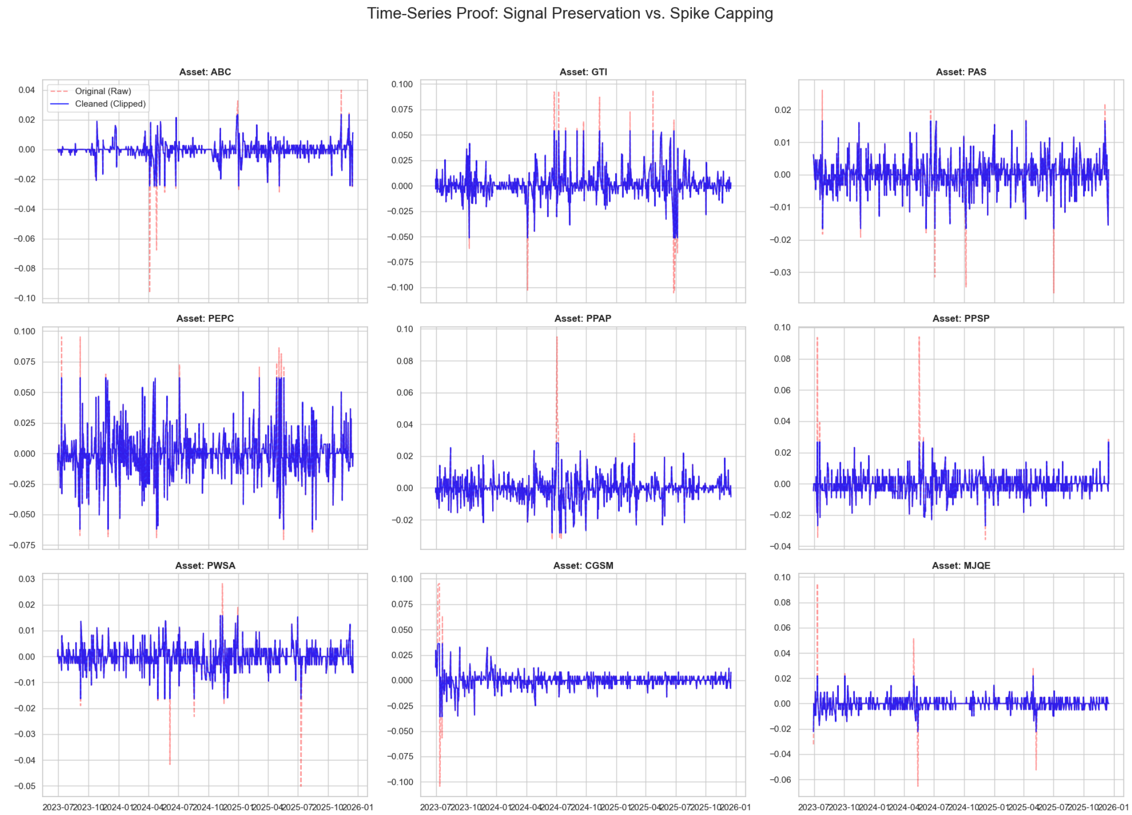

Figure 3: Time-series comparison of original and winsorized log returns. Extreme spikes are capped while the underlying return dynamics are preserved, confirming that winsorization reduces noise without altering core signals.

The boxplots show that winsorization substantially reduces extreme tails in the return distributions, improving adherence to the normality assumption. The time-series plots confirm that this adjustment preserves the underlying return dynamics while preventing isolated spikes from dominating risk estimation. This stabilization is essential for reliable covariance estimation and robust portfolio optimization under the Markowitz framework.

4.1.4 Train/Test Split for Backtesting

After cleaning the return data and stabilizing extreme observations, a key question arises: **How reliable are the estimated expected returns and risk measures when applied to future market conditions?** Estimating portfolio inputs from historical data assumes that past behavior contains useful information about the future. However, using the same data for both estimation and evaluation may lead to overly optimistic conclusions that do not reflect real investment performance.

To address this issue, this study adopts an explicit **train/test split** and conducts an **out-of-sample backtesting** analysis. The full return dataset is divided chronologically into two non-overlapping subsamples. The *in-sample period*, spanning June 2023 to April 2025 and representing approximately 70% of the observations, is used exclusively to estimate the expected return vector and covariance matrix and to construct optimal portfolios under the Markowitz framework.

The remaining data, covering May 2025 to December 2025 (approximately 30%), form the *out-of-sample period*. Portfolio weights obtained from the in-sample optimization are held fixed and applied to realized returns, with no re-estimation or re-optimization. This ensures that performance evaluation reflects genuine predictive behavior rather than in-sample fitting.

The 70%–30% split balances the need for stable parameter estimation with the ability to evaluate portfolio performance under new market conditions. This back-testing framework enhances the empirical credibility of the analysis and provides a more realistic assessment of the Markowitz model's performance in the Cambodian equity market.

4.2 Summary Statistics (In-Sample Period)

This subsection summarizes the statistical properties of daily logarithmic returns during the *in-sample period* (June 2023 to April 2025), which is used for estimating model parameters and constructing optimal portfolios. All statistics are computed exclusively from in-sample data to avoid information leakage from future observations.

Table 5: In-Sample Summary Statistics of Daily Logarithmic Returns

Asset	Count	Mean	Std.	Min	25%	50%	75%	Max	Skew	Kurt.	Ann. Ret.	Ann. Vol.
ABC	454	-0.00048	0.00612	-0.02415	-0.00261	0.00000	0.00000	0.02311	-0.577	5.581	-0.121	0.097
CGSM	454	-0.00012	0.00874	-0.03620	-0.00408	0.00000	0.00411	0.03664	0.222	6.743	-0.031	0.139
GTI	454	0.00156	0.01372	-0.05138	-0.00373	0.00000	0.00438	0.05443	1.047	4.885	0.392	0.218
MJQE	454	-0.00012	0.00527	-0.02244	-0.00456	0.00000	0.00000	0.02218	0.330	3.949	-0.031	0.084
PAS	454	-0.00009	0.00498	-0.01653	-0.00177	0.00000	0.00175	0.01656	0.099	2.193	-0.022	0.079
PEPC	454	-0.00029	0.01917	-0.06232	-0.00837	0.00000	0.00774	0.06261	0.461	2.359	-0.072	0.304
PPAP	454	-0.00033	0.00853	-0.02866	-0.00423	0.00000	0.00290	0.02868	0.180	2.408	-0.083	0.135
PPSP	454	-0.00036	0.00764	-0.02733	-0.00459	0.00000	0.00456	0.02732	0.223	2.218	-0.090	0.121
PWSA	454	-0.00031	0.00476	-0.01624	-0.00284	0.00000	0.00275	0.01573	-0.022	2.205	-0.079	0.076

Table 5 reports key descriptive measures, including central tendency, dispersion, and higher-order moments. Mean daily returns are close to zero for most assets, which is typical of short-horizon equity returns. When annualized, GTI exhibits the highest expected return, while most other stocks display slightly negative annualized returns.

Volatility varies substantially across assets. PWSA and PAS show relatively low annualized volatility, indicating more stable return behavior, whereas PEPC and GTI are considerably more volatile. These differences are economically relevant, as low-volatility assets tend to dominate minimum variance portfolios.

Skewness and kurtosis reveal notable departures from normality. Several assets exhibit asymmetric return distributions and excess kurtosis, reflecting heavy tails and occasional extreme price movements. These characteristics are consistent with the structure of an emerging market and help explain the importance of careful risk estimation prior to portfolio optimization.

5 Exploratory Data Analysis (EDA)

This section presents an exploratory analysis of the CSX-listed stocks prior to portfolio optimization. The objective is to understand price behavior, return distributions, and risk characteristics of individual assets. These insights provide economic intuition and motivate the use of mean–variance portfolio theory.

5.1 Price Performance on a Normalized Scale

To compare price movements across assets with different price levels, historical prices are rebased to a common starting value of 100. This normalization allows relative performance to be analyzed without being affected by nominal price differences.

Figure 4 shows the rebased price paths for each asset individually, while Figure 5 presents a combined comparison. Most CSX stocks exhibit gradual price movements over time, with long flat segments indicating low trading frequency. These patterns are consistent with thin liquidity and slow price adjustment in an emerging equity market.

Although short-term fluctuations are limited, the cumulative price paths diverge over time, suggesting meaningful differences in long-run performance across assets. This motivates the use of diversification to balance assets with different growth profiles.

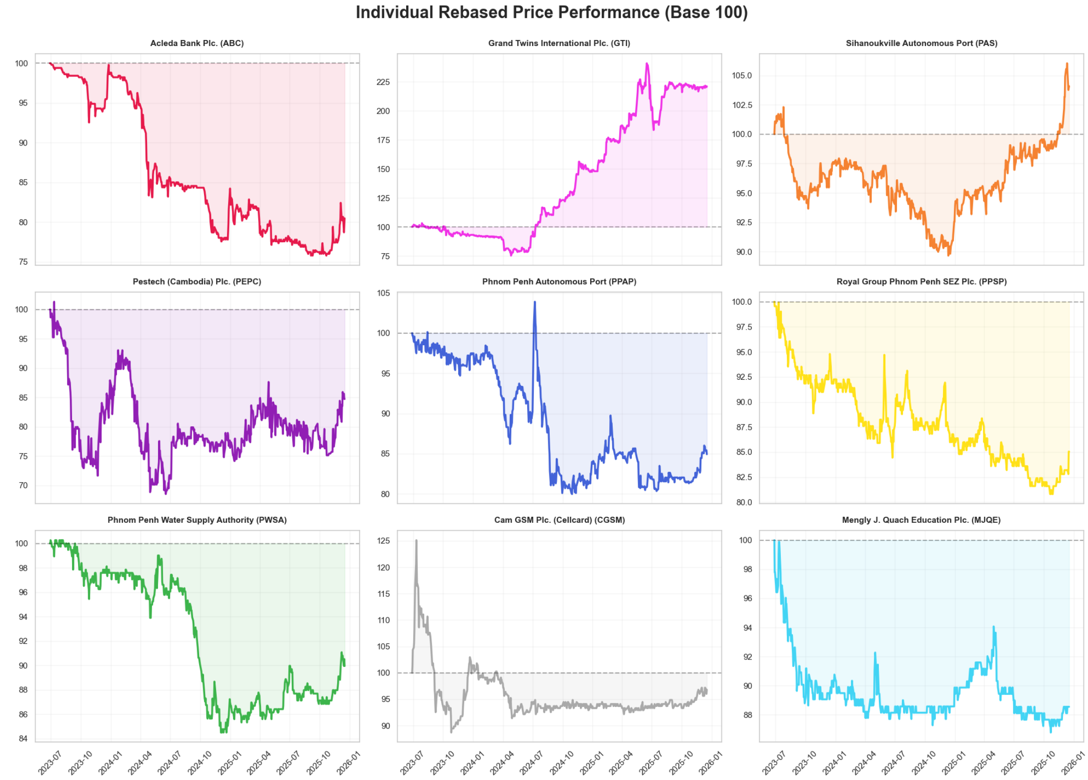

Figure 4: Individual rebased price performance of CSX-listed stocks (Base = 100).

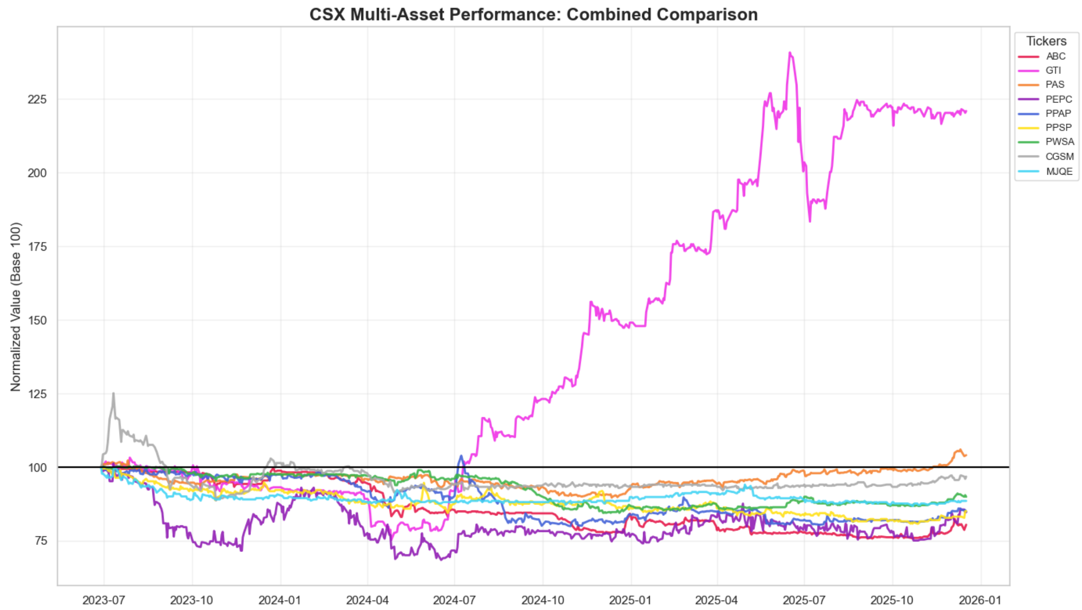

Figure 5: Combined normalized price performance of all CSX-listed stocks.

5.2 Returns Analysis and Distributional Properties

Since asset prices are non-stationary, further analysis is conducted using daily logarithmic returns. Log returns stabilize the variance and allow return distributions to be examined using standard statistical tools.

Descriptive statistics show that average daily returns are close to zero for most assets, reflecting modest price changes over short horizons. However, higher moments of the distributions reveal important features. Several stocks exhibit positive kurtosis and noticeable skewness, indicating deviations from the normal distribution.

Figure 6 compares empirical return distributions with a fitted normal curve. While the central mass of returns aligns reasonably well with normality, heavier tails are present for some assets. This behavior implies that extreme returns occur more frequently than predicted by a normal distribution, which has important implications for risk estimation.

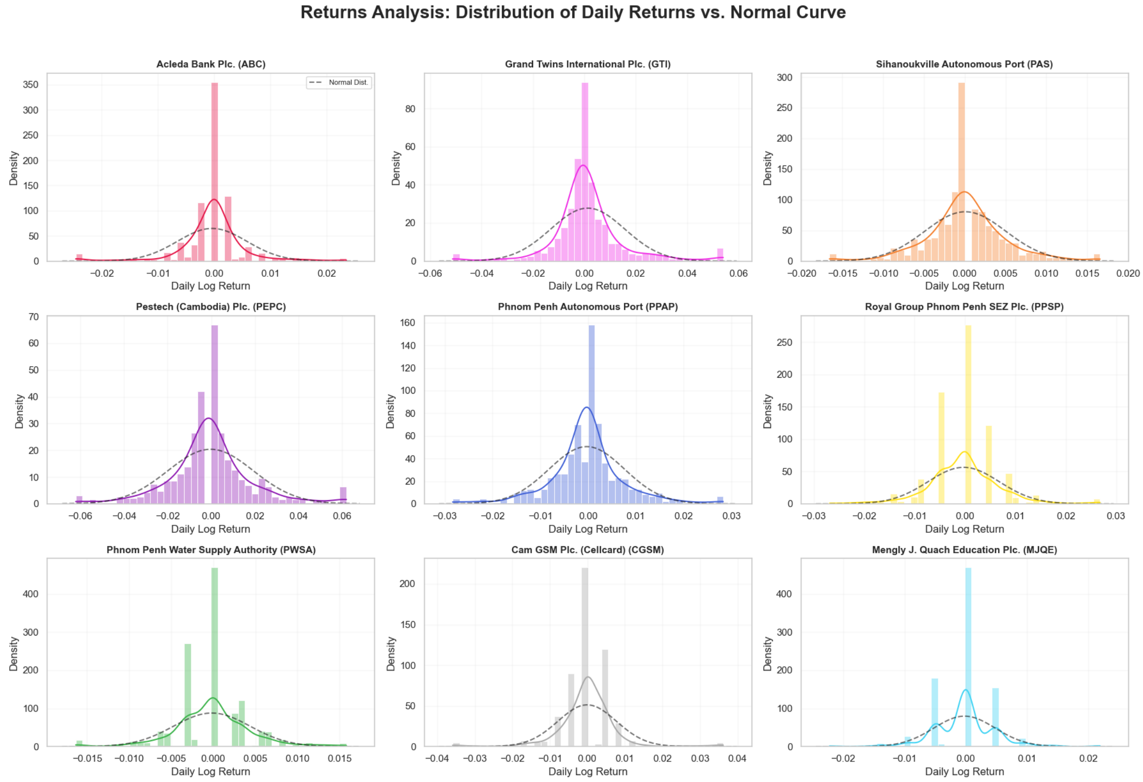

Figure 6: Distribution of daily logarithmic returns with normal density overlays.

5.3 Volatility Profile and Risk Ranking

Asset risk is measured using the annualized standard deviation of daily logarithmic returns. To ensure consistency with portfolio risk calculations, volatility is computed from the covariance matrix by assigning a full weight to each asset individually.

Figure 7 ranks assets from lowest to highest risk. Stocks such as PWSA and PAS display relatively low volatility, indicating more stable return behavior. In contrast, PEPC and GTI exhibit higher volatility, reflecting larger day-to-day price fluctuations. Overall, PEPC has the highest Volatility at 30.42 %, while PWSA has the lowest at 7.56%.

These differences are economically important for portfolio construction. Low-volatility assets tend to stabilize portfolio risk, while high-volatility assets offer greater return potential at the cost of increased risk. The observed variation in volatility across CSX stocks provides a strong motivation for mean–variance optimization.

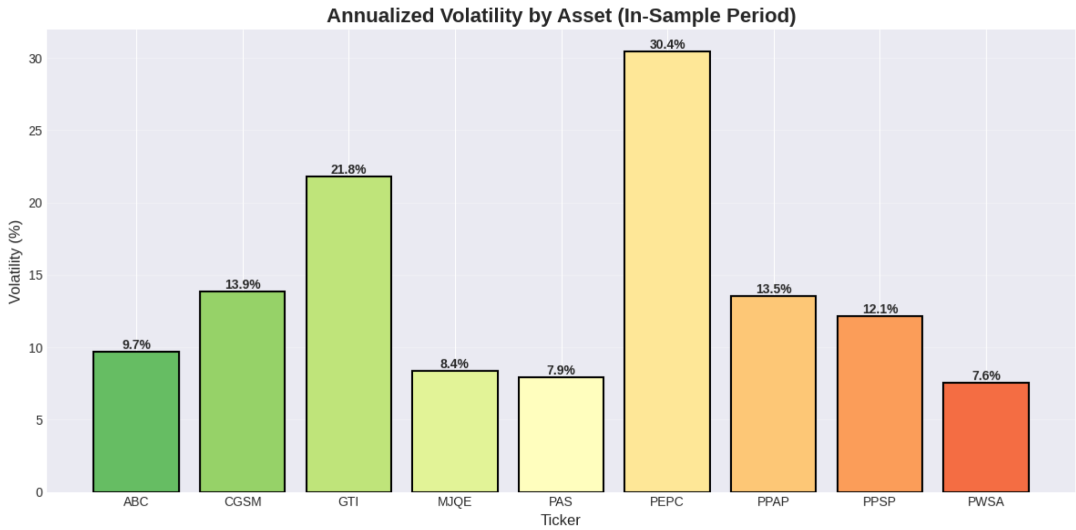

Figure 7: Annualized volatility ranking of CSX-listed stocks (cleaned returns).

5.3.1 Returns Distribution Analysis

It is very important to examine the distributional properties of daily logarithmic returns to assess the validity of the normality assumption underlying the Markowitz mean–variance framework. Normality is important because portfolio risk is summarized solely by the mean and variance of returns.

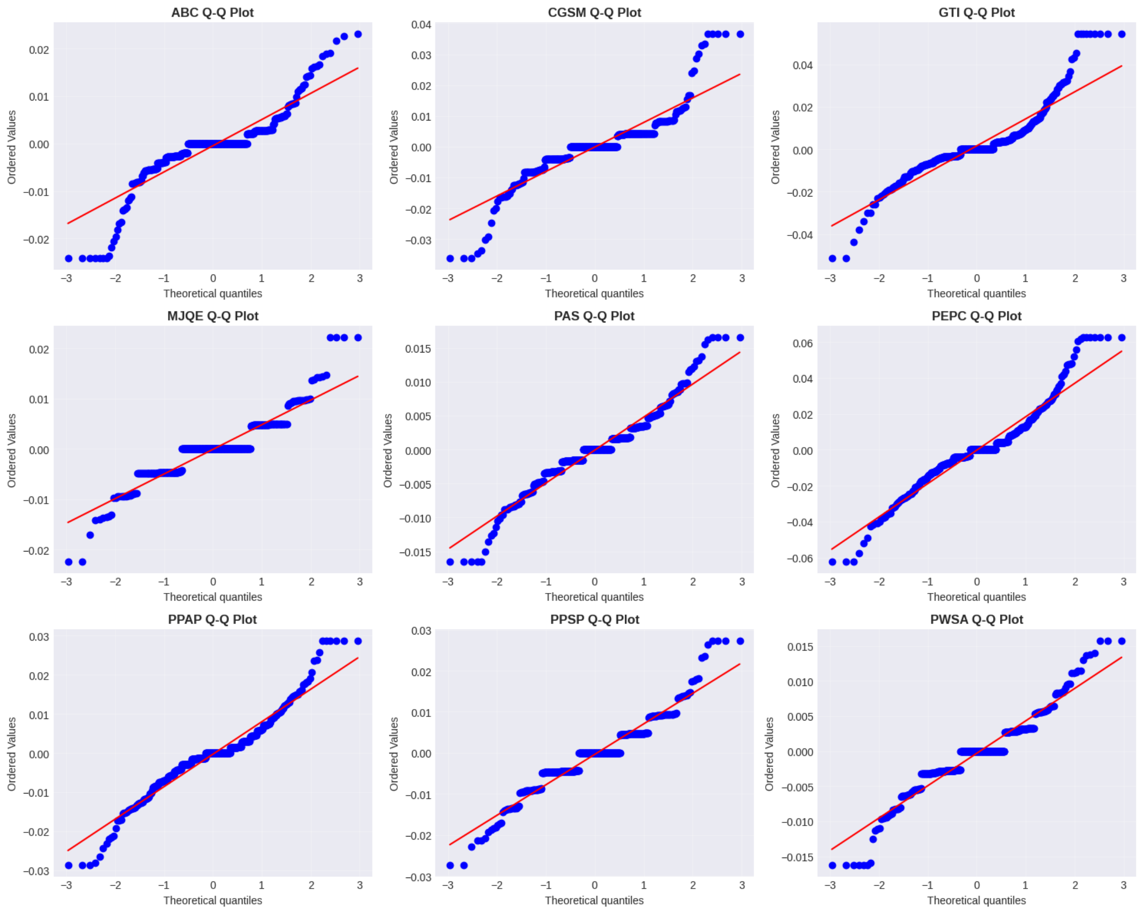

Figure 8: Q–Q plots of daily logarithmic returns for CSX-listed stocks. Deviations from the 45-degree line indicate departures from normality, particularly in the tails of the distributions.

The Q–Q plots show that the central portions of the return distributions are reasonably well aligned with the normal reference line, suggesting approximate normality around the mean. However, visible deviations at the lower and upper quantiles indicate skewness and excess kurtosis, reflecting occasional extreme price movements.

Overall, while the return distributions are not perfectly normal, they are sufficiently close in the center to justify the use of mean–variance optimization. The observed tail behavior further motivates the application of outlier treatment to stabilize risk and covariance estimation.

5.4 Risk and Correlation Framework

Portfolio risk is not determined solely by the volatility of individual assets, but also by how asset returns move relative to one another. This section examines the joint behavior of CSX-listed stocks using the covariance matrix and the correlation matrix, which together form the core inputs of the mean–variance optimization model.

5.4.1 The Covariance Matrix

The covariance matrix summarizes how pairs of asset returns co-move over time. For two assets i and j, covariance is defined as

$$\mathrm{Cov}(R_i, R_j) = \mathbb{E}\Big[(R_i - \mu_i)(R_j - \mu_j)\Big],$$

where R_i and R_j denote asset returns and μ_i, μ_j their expected values.

In this study, the covariance matrix is computed from daily logarithmic returns and then **annualized** by multiplying by 252, the approximate number of trading days in a year:

$$\Sigma_{\text{annual}} = 252\,\Sigma_{\text{daily}}.$$

Annualization converts daily risk measures into an annual scale, which is more appropriate for portfolio evaluation and optimization.

The diagonal elements of the covariance matrix represent individual asset variances, while the off-diagonal elements capture cross-asset dependence. Lower covariances indicate stronger diversification potential, as losses in one asset are less likely to coincide with losses in another.

Table 6: Annualized covariance matrix of CSX stock returns (in-sample period). Diagonal elements represent asset variances, while off-diagonal elements measure comovements between asset returns.

	ABC	CGSM	GTI	MJQE	PAS	PEPC	PPAP	PPSP	PWSA
ABC	0.009451	0.001249	0.003166	0.001582	0.000732	0.000682	0.001241	0.001792	0.001656
CGSM	0.001249	0.019235	0.000902	0.001576	0.000898	0.006421	0.000470	0.002699	0.000660
GTI	0.003166	0.000902	0.047425	0.001753	0.000240	-0.000414	-0.000206	0.003301	0.002320
MJQE	0.001582	0.001576	0.001753	0.006984	0.000642	0.001597	0.000154	0.000420	0.000172
PAS	0.000732	0.000898	0.000240	0.000642	0.006238	0.001435	0.000723	0.000092	0.001220
PEPC	0.000682	0.006421	-0.000414	0.001597	0.001435	0.092564	0.001502	0.004757	0.002637
PPAP	0.001241	0.000470	-0.000206	0.000154	0.000723	0.001502	0.018354	0.001849	0.000483
PPSP	0.001792	0.002699	0.003301	0.000420	0.000092	0.004757	0.001849	0.014706	-0.000189
PWSA	0.001656	0.000660	0.002320	0.000172	0.001220	0.002637	0.000483	-0.000189	0.005718

5.4.2 Correlation Heatmap

While covariance measures joint variability in absolute terms, correlation standardizes this relationship to lie between −1 and 1, making comparisons across asset pairs easier. The correlation coefficient is defined as

$$\rho_{ij} = \frac{\mathrm{Cov}(R_i, R_j)}{\sigma_i \sigma_j},$$

where σ_i and σ_j are the standard deviations of assets i and j.

The correlation heatmap provides a visual summary of these relationships. Values close to +1 indicate strong positive co-movement, values near 0 suggest weak dependence, and negative values imply that assets tend to move in opposite directions.

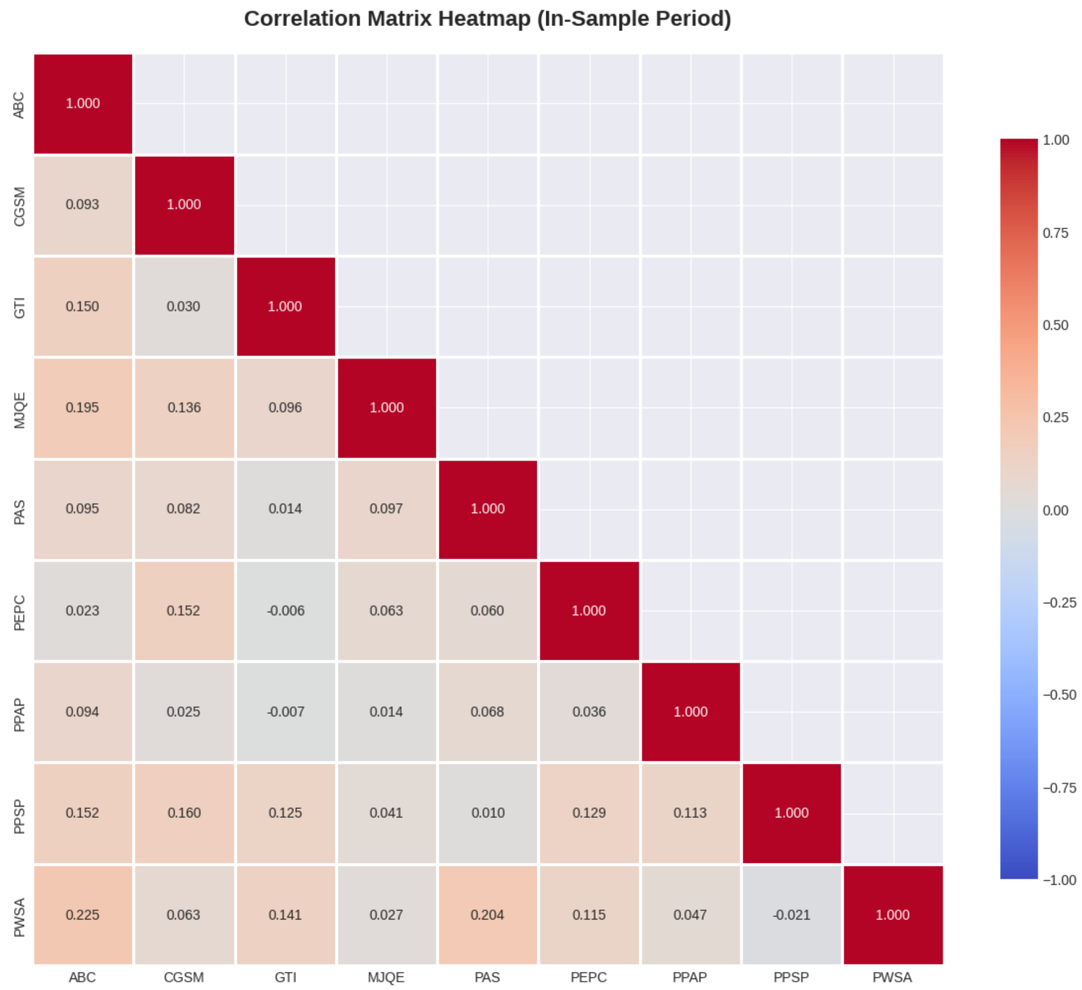

Figure 9: Correlation heatmap of CSX stock returns. Red tones indicate high positive correlation, while blue tones indicate low or negative correlation.

The correlation heatmap confirms that correlations across CSX stocks are mostly weak to moderate, with several pairs exhibiting near-zero or slightly negative correlation. This indicates meaningful diversification potential within the market. In particular, asset pairs such as PEPC–GTI and PPAP–GTI display the lowest correlations, implying that holding these assets together can reduce portfolio risk more effectively than combining highly correlated stocks.

5.4.3 Why Correlation Matters for Diversification

Correlation plays a central role in portfolio construction. If two stocks are highly correlated, they tend to rise and fall together, providing little diversification benefit when held jointly. In contrast, combining assets with low or negative correlation can reduce overall portfolio risk without lowering expected return.

The correlation structure observed in the CSX data shows generally low to moderate correlations across most asset pairs. This suggests that meaningful diversification benefits are available, even within a relatively small and illiquid market. These properties are later exploited in the construction of minimum variance and maximum Sharpe ratio portfolios.

6 Portfolio Optimization Methodology

This study applies the framework of Modern Portfolio Theory (MPT), originally developed by Markowitz, to determine optimal asset allocations for CSX-listed stocks. The core idea of MPT is that portfolio risk depends not only on the risk of individual assets, but also on how their returns co-move. By combining assets with imperfect correlations, investors can reduce total risk without necessarily reducing expected return.

6.1 Mean–Variance Optimization Framework

Under the mean–variance framework, a portfolio is fully characterized by its expected return and variance. Let $\mathbf{w}$ denote the vector of portfolio weights, μ the vector of expected asset returns, and Σ the covariance matrix of returns. Portfolio risk is measured by the quadratic form $\mathbf{w}^\top \Sigma \mathbf{w}$.

Two optimization environments are considered:

1. **With short selling allowed**
2. **Without short selling (long-only)**

Each environment is examined using analytical and numerical methods where applicable.

6.2 Case 1: Portfolio Optimization with Short Selling Allowed

6.2.1 Global Minimum Variance Portfolio (GMVP)

The Global Minimum Variance Portfolio (GMVP) represents the portfolio with the lowest attainable risk among all fully invested portfolios, independent of any target expected return. It serves as the natural starting point in mean–variance portfolio theory, as it depends solely on the covariance structure of asset returns. In this study, the GMVP is constructed under two different market settings, depending on whether short selling is allowed. Although the objective function is identical in both cases, the nature of the constraints fundamentally alters the solution method.

Analytical Solution (Closed-Form). When short selling is permitted, the optimization problem includes only an equality constraint requiring full investment. No restrictions are imposed on the sign of portfolio weights. The resulting optimization problem admits a closed-form analytical solution derived using Lagrange multipliers.

Formally, the problem is:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1$$

Under this unconstrained setting, the optimal portfolio weights are given by:

$$\mathbf{w}_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}$$

This solution depends only on the covariance matrix and ignores expected returns entirely. Because no inequality constraints are imposed, some weights may be negative, implying short positions. From a mathematical perspective, this outcome is natural: allowing negative weights enables the portfolio to hedge correlated risks more effectively, thereby achieving the absolute minimum variance.

Empirical Results.

Table 7 reports the estimated weights of the Global Minimum Variance Portfolio when short selling is allowed. The portfolio achieves an annualized **minimum risk of 4.4461%** with an **expected return of -5.4771%**, while satisfying the full-investment condition.

The resulting allocation is dominated by relatively low-volatility assets, with the largest weights assigned to PWSA, MJQE, and PAS. A small negative weight is allocated to PEPC, confirming that short selling is actively used by the optimizer to hedge

Table 7: Global Minimum Variance Portfolio Weights (Short Selling Allowed)

Ticker	**Asset Name**	**Weight (%)**
PWSA	Phnom Penh Water Supply Authority	26.23
MJQE	Mengly J. Quach Education Plc.	22.12
PAS	Sihanoukville Autonomous Port	21.82
PPSP	Royal Group Phnom Penh SEZ Plc.	10.32
PPAP	Phnom Penh Autonomous Port	7.43
ABC	ACLEDA Bank Plc.	7.19
CGSM	Cambodia GSM Plc. (Cellcard)	4.53
GTI	Grand Twins International Plc.	0.70
PEPC	Pestech (Cambodia) Plc.	−0.35

portfolio risk. Overall, the composition reflects a strong preference for stability, as expected from a variance-minimizing objective, while illustrating how unconstrained optimization naturally incorporates short positions to achieve the lowest possible risk.

Numerical Verification (SLSQP). The GMVP is re-estimated using Sequential Least Squares Programming (SLSQP). This is to ensure that the closed-form form is correct.

Table 8: GMVP Weights (Numerical SLSQP, Short Selling Allowed)

Ticker	Asset Name	Weight (%)
PWSA	Phnom Penh Water Supply Authority	26.23
MJQE	Mengly J. Quach Education Plc.	22.13
PAS	Sihanoukville Autonomous Port	21.82
PPSP	Royal Group Phnom Penh SEZ Plc.	10.34
PPAP	Phnom Penh Autonomous Port	7.42
ABC	ACLEDA Bank Plc.	7.19
CGSM	Cambodia GSM Plc.	4.53
GTI	Grand Twins International Plc.	0.70
PEPC	Pestech (Cambodia) Plc.	−0.35

Table 9: Analytical vs. Numerical GMVP Comparison

Ticker	Analytical	Numerical	Difference	Abs. Diff
ABC	0.071856	0.071854	0.000002	0.000002
CGSM	0.045328	0.045281	0.000047	0.000047
GTI	0.007020	0.007001	0.000018	0.000018
MJQE	0.221186	0.221265	-0.000078	0.000078
PAS	0.218231	0.218191	0.000040	0.000040
PEPC	-0.003471	-0.003477	0.000006	0.000006
PPAP	0.074291	0.074193	0.000098	0.000098
PPSP	0.103228	0.103440	-0.000212	0.000212
PWSA	0.262331	0.262252	0.000079	0.000079

From the table, it is observable that the errors are very small which indicates that our closed-form formula calculated the weights correctly.

6.2.2 Minimum Variance Line with Short Selling Allowed

While the Global Minimum Variance Portfolio identifies a single portfolio with the lowest attainable risk, investors are often interested in understanding how risk changes when a specific level of expected return is required. This leads to the concept of the *minimum variance line*, which represents the set of portfolios that minimize variance for each admissible target return μ_V.

When short selling is allowed, the optimization problem involves only equality constraints and therefore admits a closed-form analytical solution. For a given target expected return μ_V, the problem is formulated as:

$$\mathbf{w} = \frac{\begin{vmatrix} 1 & \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{m} \\ \mu_V & \mathbf{m}^\top \mathbf{C}^{-1}\mathbf{m} \end{vmatrix} \mathbf{C}^{-1}\mathbf{u} + \begin{vmatrix} \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{u} & 1 \\ \mathbf{m}^\top \mathbf{C}^{-1}\mathbf{u} & \mu_V \end{vmatrix} \mathbf{C}^{-1}\mathbf{m}}{\begin{vmatrix} \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{u} & \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{m} \\ \mathbf{m}^\top \mathbf{C}^{-1}\mathbf{u} & \mathbf{m}^\top \mathbf{C}^{-1}\mathbf{m} \end{vmatrix}}$$

Let:

$$A = \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{u}, \quad B = \mathbf{u}^\top \mathbf{C}^{-1}\mathbf{m}, \quad C = \mathbf{m}^\top \mathbf{C}^{-1}\mathbf{m}, \quad D = AC - B^2$$

Then each determinant equals a scalar multiple of the numerator/denominator used in the derivation:

- **Denominator:** $\begin{vmatrix} A & B \\ B & C \end{vmatrix} = AC - B^2 = D$

- **First numerator determinant:** $\begin{vmatrix} 1 & B \\ \mu_V & C \end{vmatrix} = C - \mu_V B$
- **Second numerator determinant:** $\begin{vmatrix} A & 1 \\ B & \mu_V \end{vmatrix} = A\mu_V - B$

Plugging these into the original expression and factoring out $\mathbf{C}^{-1}$ gives:

$$\mathbf{w}(\mu_V) = \mathbf{C}^{-1}\left[\frac{C - \mu_V B}{D}\mathbf{u} + \frac{\mu_V A - B}{D}\mathbf{m}\right]$$

This expression shows that portfolio weights depend linearly on the target return μ_V. Varying μ_V over its admissible range generates a continuum of optimal portfolios, which together form the minimum variance line.

Efficient and Inefficient Sets.

The minimum variance line is symmetric around the Global Minimum Variance Portfolio. Portfolios located below the GMVP deliver lower expected return for higher or equal risk and are therefore inefficient. In contrast, the upper branch of the curve represents the *efficient frontier*, consisting of portfolios that offer the highest expected return for a given level of risk.

Empirical Illustration.

Figure 10 illustrates the complete minimum variance line obtained under short-selling conditions, including both the efficient and inefficient sets. Individual assets are shown for reference, and the GMVP appears as the vertex of the hyperbola:

$$\sigma^2(\mu) = \frac{505.8733\,\mu^2 - 55.4145\,\mu + 7.5370}{3045.0886}$$

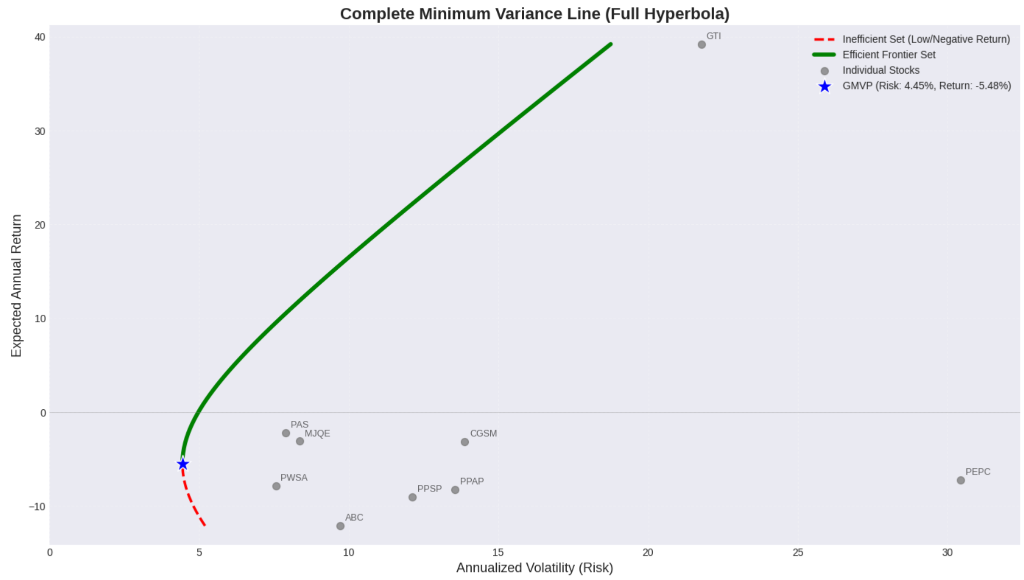

Figure 10: Minimum variance line with short selling allowed. The lower branch represents inefficient portfolios, while the upper branch corresponds to the efficient frontier. The Global Minimum Variance Portfolio is located at the vertex of the curve.

6.3 Case 2: Portfolio Optimization without Short Selling (Long-Only)

6.3.1 Global Minimum Variance Portfolio (Long-Only)

When short selling is prohibited, portfolio weights are restricted to be non-negative. This introduces inequality constraints that prevent the use of closed-form analytical solutions. Consequently, the Global Minimum Variance Portfolio must be obtained numerically.

The constrained optimization problem is given by

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \sum_{i=1}^{n} w_i = 1, \qquad 0 \le w_i \le 1,\ i = 1, \dots, n.$$

Because of the long-only constraint, the feasible set is restricted to the unit simplex. In this study, the problem is solved using Sequential Least Squares Programming (SLSQP), which is well suited for quadratic objective functions with both equality and inequality constraints.

Table 10: Global Minimum Variance Portfolio Weights (Long-Only, No Short Selling)

Ticker	Weight	Weight (%)
ABC	0.072300	7.23
CGSM	0.044502	4.45
GTI	0.007233	0.72
MJQE	0.220825	22.08
PAS	0.217981	21.80
PEPC	0.000000	0.00
PPAP	0.074240	7.42
PPSP	0.102080	10.21
PWSA	0.260838	26.08
Total	**1.000000**	**100.00**

The long-only GMV portfolio achieves an expected annualized return of −5.4725% with a volatility of 4.4473%. All portfolio weights satisfy the non-negativity constraint, with PEPC receiving zero allocation under the long-only restriction.

6.3.2 Minimum Variance Line (Long-Only)

When short selling is prohibited, the construction of the minimum variance line requires a numerical approach. In contrast to the unconstrained case, portfolio weights are restricted to be non-negative and bounded above, which introduces inequality constraints into the optimization problem. These constraints prevent the use of closed-form analytical solutions and necessitate the use of constrained numerical optimization.

For a given target expected return μ_V, the minimum variance portfolio is obtained by solving

$$Minimize(Variance) = \min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$$

Subject to:

$$\begin{cases} \mathbf{w}^\top \mu = \mu_V \\ \mathbf{1}^\top \mathbf{w} = 1 \end{cases}$$

Short selling is not allowed and position sizes are limited:

$$0 \leq w_i \leq 1, \quad i = 1, \ldots, n$$

The feasible set is therefore given by

$$\mathcal{W} = \left\{ \mathbf{w} \in \mathbb{R}^n : \sum_{i=1}^{n} w_i = 1,\ 0 \leq w_i \leq 1 \right\},$$

which corresponds to fully invested, long-only portfolios with limited position sizes.

Because of the inequality constraints on the portfolio weights, the optimization problem becomes a quadratic programming problem. In this study, it is solved numerically using Sequential Least Squares Programming (SLSQP). For each target expected return μ_V, the algorithm iteratively searches for the portfolio that achieves the lowest possible variance while satisfying all constraints. Repeating this procedure over a range of target returns generates a sequence of optimal portfolios that together form the numerical minimum variance line.

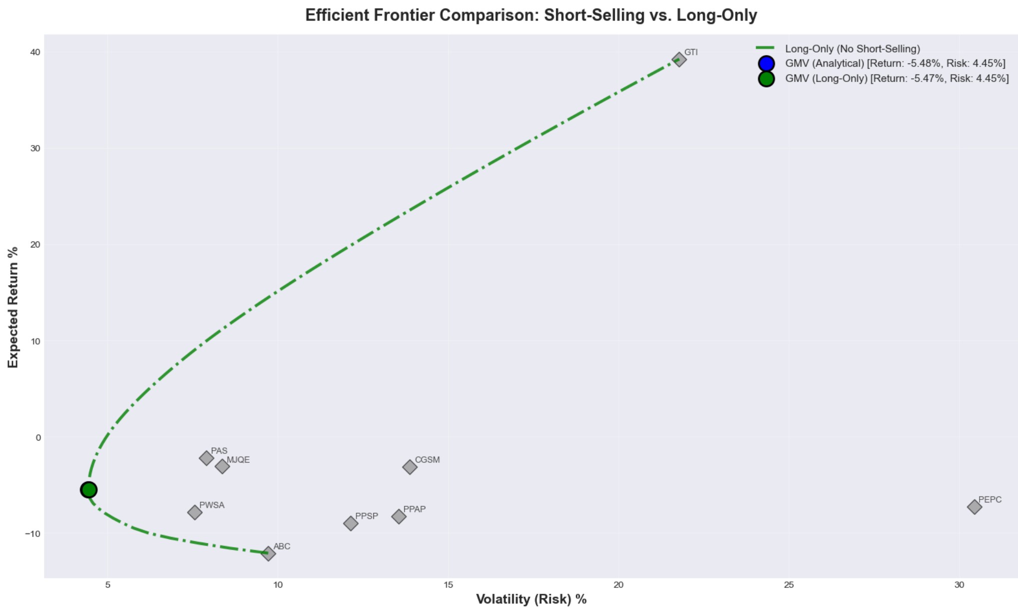

Figure 11: Numerical minimum variance line and efficient frontier under a no short-selling constraint. The star denotes the global minimum variance portfolio, individual assets are shown as points, and the upper segment of the curve corresponds to the efficient frontier.

Figure 11 illustrates the resulting minimum variance line. As in the theoretical framework, the global minimum variance portfolio appears at the vertex of the curve. Portfolios below this point are inefficient, while portfolios above it form the efficient frontier. Compared to the unconstrained case, the frontier is narrower and less smooth, reflecting the loss of flexibility caused by prohibiting short positions.

6.4 Comparative Analysis

6.4.1 Visual Comparison of Efficient Frontiers

Figure 12 overlays all efficient frontiers to provide a direct visual comparison between optimization methods and constraint settings. Specifically, it combines the analytical and numerical efficient frontiers with short selling allowed, alongside the numerical long-only efficient frontier. Individual assets and the corresponding Global Minimum Variance (GMV) portfolios are also shown for reference.

The analytical and numerical frontiers under short-selling conditions overlap almost perfectly, confirming that the numerical SLSQP optimizer accurately replicates the closed-form theoretical solution. This close alignment validates the correctness of both implementations.

In contrast, the long-only frontier is shifted inward. For any given level of expected return, the long-only portfolio exhibits higher risk, reflecting the loss of diversification flexibility when short selling is prohibited. The long-only GMV portfolio also appears at a slightly higher risk level than its unconstrained counterpart, illustrating the cost of imposing non-negativity constraints on portfolio weights.

Overall, the figure highlights two key insights: short selling expands the efficient frontier and improves attainable risk–return trade-offs, while long-only constraints reduce efficiency but yield more practically implementable portfolios.

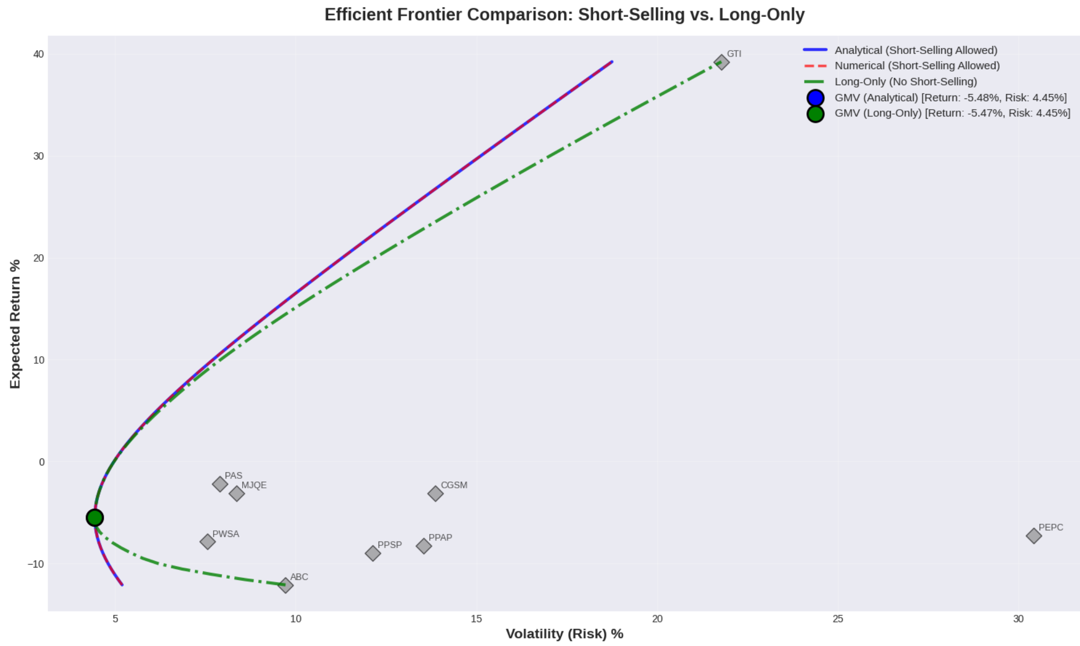

Figure 12: Overlay of efficient frontiers under short-selling and long-only constraints. The analytical and numerical frontiers with short selling allowed nearly coincide, while the long-only frontier is shifted inward, indicating higher risk for comparable expected returns. GMV portfolios and individual assets are shown for reference.

6.4.2 Short-Selling Impact Analysis

This subsection examines how short-selling constraints affect portfolio performance by comparing the Global Minimum Variance (GMV) portfolio with and without short-selling permission.

Table 11: Impact of Short-Selling Constraints on the GMV Portfolio

Metric	Short-Selling Allowed	Long-Only
Expected Return (%)	−5.48	−5.47
Volatility (%)	4.45	4.45
Negative Weights	1 asset	0 assets
Zero Weights	–	1 asset
Δ Return (%)	+0.005	
Δ Volatility (%)	+0.001	

The results indicate that allowing short selling leads to a marginally lower portfolio risk by enabling limited hedging through negative positions. When short selling is prohibited, one asset weight becomes binding at zero, slightly reducing diversification and increasing portfolio volatility.

Although the numerical differences at the GMV point are economically small, the table highlights a key structural effect: short-selling constraints restrict the feasible set of portfolios. This effect becomes more pronounced away from the GMV point, as reflected in the inward shift and reduced smoothness of the long-only efficient frontier observed in the previous visual comparison.

6.4.3 Interpretation and Practical Implications

The comparative analysis of minimum variance lines highlights the economic and practical consequences of imposing short-selling constraints in portfolio optimization.

First, the shape and position of the efficient frontier are directly affected by the constraint. The long-only minimum variance line lies entirely to the right of the unconstrained frontier, indicating that portfolios constructed without short selling are dominated in a mean–variance sense. For a given level of expected return, long-only portfolios exhibit higher risk, while for a given level of risk they offer lower expected returns.

Second, the long-only constraint alters the risk–return trade-off faced by investors. The inability to take short positions limits diversification opportunities and prevents the use of hedging strategies. As a result, the efficient frontier becomes narrower and less smooth, reflecting the presence of binding constraints and corner solutions.

Third, portfolio composition is materially affected. Long-only optimization tends to produce more concentrated portfolios, with several assets receiving zero weight as target returns increase. This contrasts with the unconstrained case, where negative weights allow risk to be distributed more evenly across assets.

From a practical perspective, these findings must be interpreted in light of real-world investment constraints. Short selling is often unavailable or restricted for retail investors, and even when permitted, it involves borrowing costs, margin requirements, and additional transaction costs. Moreover, regulatory and institutional constraints frequently prohibit short positions altogether.

Consequently, while portfolios that allow short selling are theoretically more efficient, long-only portfolios remain more realistic and easier to implement in practice. For most individual investors, the simplicity, transparency, and lower operational risk of long-only strategies may outweigh the modest loss in theoretical efficiency identified in this study.

7 Optimal Portfolio Weights and Asset Allocation

Having examined the efficient frontiers and the impact of short-selling constraints on the risk–return trade-off, this section focuses on the **optimal portfolio weights** implied by the Global Minimum Variance (GMV) strategy. All allocations reported in this section are based exclusively on **in-sample optimization**, ensuring that portfolio construction does not use any out-of-sample information.

The GMV portfolio is of particular interest because it represents the most risk-efficient allocation achievable under a given set of constraints. To assess the robustness of the resulting asset allocation, portfolio weights are compared across three settings: (i) the analytical closed-form solution with short selling allowed, (ii) the numerical SLSQP solution under the same conditions, and (iii) the numerical long-only solution without short selling.

Table 12: Comparison of Global Minimum Variance Portfolio Weights (%)

Ticker	GMV (Analytical)	GMV (Numerical)	GMV (Long-Only)
ABC	7.19	7.19	7.23
CGSM	4.53	4.53	4.45
GTI	0.70	0.70	0.72
MJQE	22.12	22.13	22.08
PAS	21.82	21.82	21.80
PEPC	−0.35	−0.35	0.00
PPAP	7.43	7.42	7.42
PPSP	10.32	10.34	10.21
PWSA	26.23	26.23	26.08

Table 12 shows a high degree of consistency between the analytical and numerical GM

7.1 Portfolio Performance Metrics (In-Sample)

Having identified the optimal asset allocations, it is natural to ask how these portfolios actually perform. This subsection evaluates the in-sample expected returns and volatilities of the GMV portfolios obtained under different optimization approaches.

Table 13: In-Sample Performance Metrics of GMV Portfolios

Portfolio	**Expected Return (%)**	**Volatility (%)**
GMV (Analytical)	−5.48	4.45
GMV (Numerical)	−5.48	4.45
GMV (Long-Only)	−5.47	4.45

The analytical and numerical GMV portfolios with short selling allowed exhibit virtually identical expected returns and volatilities, confirming the numerical accuracy of the SLSQP optimization. Imposing the long-only constraint results in a marginal increase in volatility and a slightly higher expected return. These differences are economically negligible, indicating that GMV performance is highly robust to short-selling constraints in-sample.

7.2 Weight Visualization: GMV Analytical vs. GMV Long-Only

While numerical tables provide precise portfolio weights, they often obscure how constraints affect the overall allocation structure. To better understand the impact of short-selling restrictions, this subsection presents a visual comparison of asset weights under the analytical Global Minimum Variance (GMV) solution and the long-only GMV solution.

The bar chart in Figure 13 compares portfolio weights obtained when short selling is allowed (analytical solution) and when it is prohibited (numerical long-only solution). Both portfolios are constructed using the same in-sample estimates of expected returns and covariances.

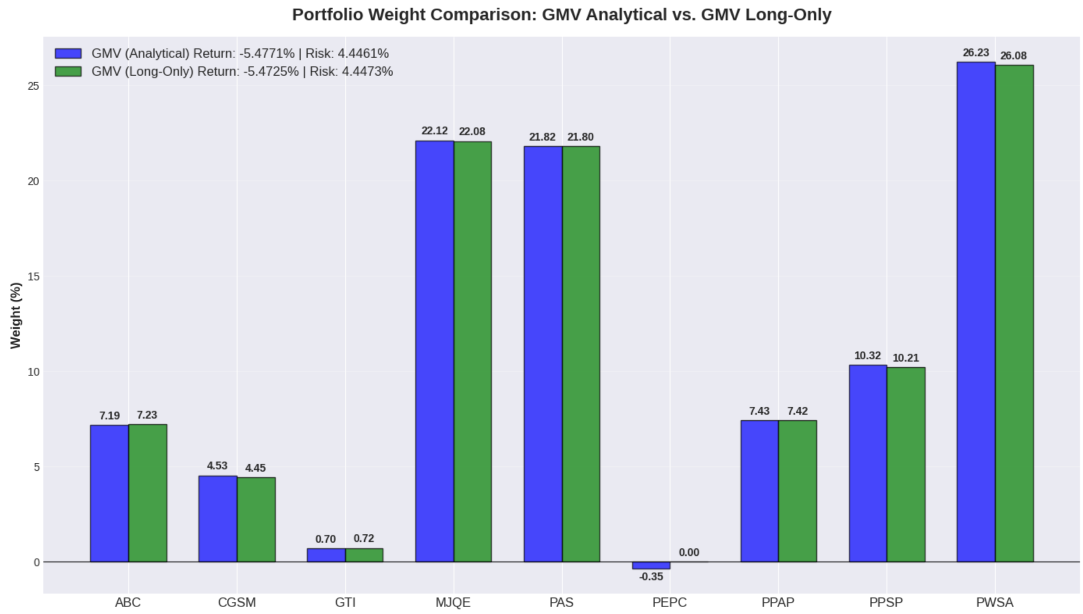

Figure 13: Portfolio weight comparison between analytical GMV (short-selling allowed) and GMV long-only portfolios.

Discussion of Portfolio Weights.

The analytical GMV portfolio permits short selling and therefore represents the theoretically minimum variance allocation. This flexibility is reflected in a small negative weight assigned to PEPC, indicating that the asset is used as a hedging component rather than a long investment. In contrast, the long-only GMV portfolio enforces the non-negativity constraint ($w_i \geq 0$), forcing PEPC's weight to zero and redistributing the remaining capital across other assets.

Despite this structural difference, the two portfolios are remarkably similar. For most assets, weight differences are well below one percentage point. The largest adjustment occurs for PEPC, where the short position in the analytical solution is removed in the long-only case. All other assets retain nearly identical allocations across both portfolios.

This similarity indicates that the unconstrained GMV solution already lies very close to the long-only feasible region. As a result, imposing short-selling constraints has only a marginal effect on portfolio composition and risk characteristics. From a practical perspective, this suggests that for the selected CSX stocks, diversification benefits are primarily achieved through long positions rather than through short-selling strategies.

Taken together, the weight comparison shows that short-selling plays only a limited role in reducing portfolio variance for the selected CSX assets. The analytical solution differs from the long-only portfolio primarily through a single small short position in PEPC, while all other asset allocations remain nearly unchanged. This indicates that minimum variance is achieved mainly through diversification across low-volatility assets rather than through aggressive hedging via short positions.

8 Out-of-Sample Backtesting & Performance Validation

8.1 Backtesting Methodology

The purpose of out-of-sample backtesting is to evaluate whether portfolios constructed using historical data remain effective when applied to new, unseen market conditions. Rather than re-optimizing portfolios using future information, this study adopts a strict separation between estimation and evaluation periods to avoid look-ahead bias.

Portfolio weights are first estimated using the *in-sample period* (June 2023 to April 2025). These weights are treated as fixed and are then applied to returns observed in the *out-of-sample period* (May 2025 to December 2025). No parameter updates or rebalancing are performed during the evaluation window, ensuring that performance reflects genuine out-of-sample behavior.

Specifically, the backtesting procedure follows four steps:

1. Compute optimal portfolio weights using in-sample returns only.
2. Apply these weights to daily out-of-sample returns.
3. Calculate realized portfolio returns and volatility over the out-of-sample period.
4. Compare realized performance with in-sample expectations to assess model robustness.

This framework allows for a direct assessment of whether the risk-minimizing properties identified during optimization translate into stable performance under different market conditions. As a result, the backtesting analysis serves as a practical validation of the Markowitz-based portfolios beyond purely in-sample results.

8.2 Benchmark Portfolio Construction (Equal-Weighted)

To evaluate whether portfolio optimization adds value, a simple benchmark is required. This study adopts an **equal-weighted portfolio** as a control strategy, where capital is distributed uniformly across all $n = 9$ CSX-listed stocks. Each asset therefore receives a fixed allocation of 11.11%, ignoring differences in risk, return, and correlation.

Table 14: In-Sample Performance Comparison: GMV vs. Equal-Weighted Benchmark

Portfolio	Expected Return (%)	Volatility (%)
Equal-Weighted Benchmark	−1.52	6.30
Global Minimum Variance (GMV)	−5.48	3.48

Despite having a higher expected return, the equal-weighted benchmark exhibits substantially greater volatility than the GMV portfolio. The optimized GMV allocation reduces portfolio risk by approximately **37%** relative to the benchmark, even though both portfolios invest in the same set of assets.

Interpretation.

The benchmark portfolio represents uninformed diversification, treating all assets as equally risky. In contrast, the GMV portfolio explicitly accounts for the covariance structure of returns and reallocates capital toward lower-risk assets. The large reduction in volatility demonstrates that the Markowitz framework delivers meaningful risk control beyond naïve allocation strategies, confirming the value of optimization in this market setting.

8.3 Out-of-Sample Performance Results

This section evaluates the out-of-sample performance of portfolios whose weights were estimated using in-sample data up to April 2025. The optimized weights are held fixed and applied to daily returns from May 2025 to December 2025, allowing for an objective assessment of model robustness under unseen market conditions.

8.3.1 Portfolio Performance Metrics – Out-of-Sample (Realized Performance)

To validate the effectiveness of the optimization framework, several realized performance metrics are computed using out-of-sample returns. These include annualized realized returns, annualized volatility, cumulative performance, and maximum drawdown. Together, these measures capture both average performance and downside risk.

Table 15 reports the realized performance metrics for the Global Minimum Variance (GMV) portfolios and the equal-weighted benchmark.

Table 15: Out-of-Sample Performance Metrics (May 2025 – December 2025)

Portfolio	Realized Return (%)	Realized Volatility (%)	Max Drawdown (%)
GMV (Analytical)	3.49	3.48	-2.04
GMV (Long-Only)	3.51	3.49	-2.05
Equal-Weighted	5.88	5.55	-3.77

The GMV portfolios achieve substantially lower realized volatility and smaller drawdowns relative to the benchmark, confirming the risk-reduction objective of the optimization framework.

8.3.2 Performance by Portfolio Type

The two GMV portfolios exhibit nearly identical realized risk–return profiles. The analytical GMV portfolio achieves the lowest realized volatility (3.48%) and the smallest maximum drawdown (-2.04%), while the long-only GMV portfolio displays only marginally higher risk.

In contrast, the equal-weighted benchmark delivers the highest realized return (5.88%) but at the cost of significantly higher volatility (5.55%) and deeper drawdowns. This highlights the classical trade-off between return maximization and risk control.

These results indicate that short-selling constraints have a limited impact on GMV performance for the CSX asset universe, while optimization itself delivers meaningful risk reduction relative to naive diversification.

8.3.3 Performance Visualization

Figure 14 illustrates the cumulative out-of-sample returns for all portfolios. The two GMV portfolios follow closely overlapping paths throughout the evaluation period, reflecting their similar volatility profiles.

The equal-weighted benchmark exhibits larger fluctuations and deeper interim losses, consistent with its higher realized volatility and maximum drawdown. Overall, the visualization confirms that GMV portfolios provide smoother and more stable performance out-of-sample.

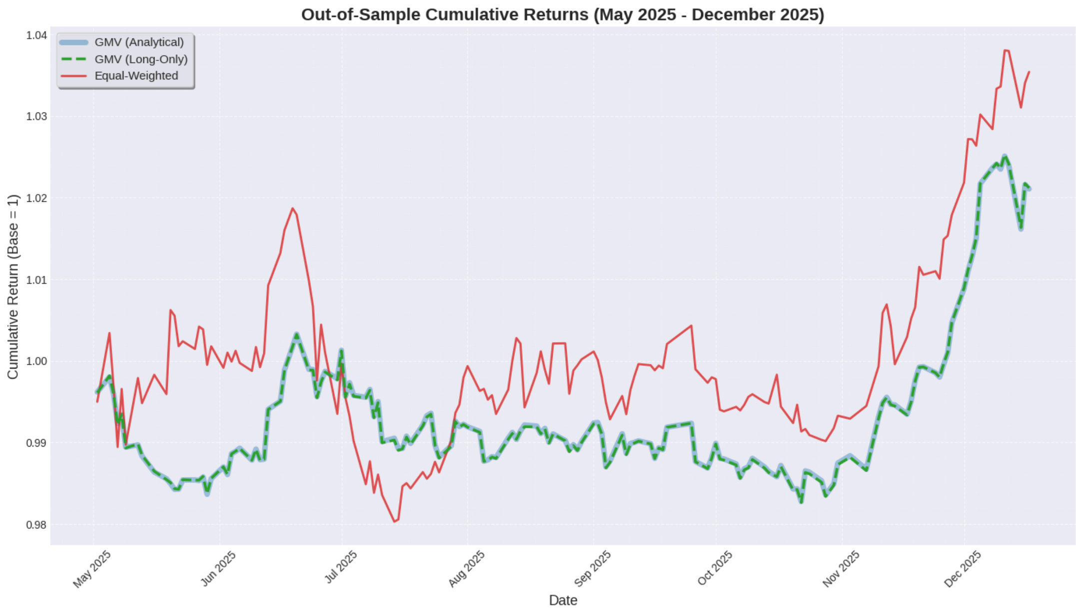

Figure 14: Out-of-sample cumulative returns of GMV and benchmark portfolios (May 2025 – December 2025).

Expected vs. Realized Performance Analysis

A central question in empirical portfolio optimization is whether performance estimated from historical data translates into realized performance under new market conditions. This section directly compares *in-sample expectations* with *out-of-sample realizations*, providing a critical evaluation of model accuracy, robustness, and practical relevance.

8.3.4 Comparison of Expected and Realized Performance

Table 16 contrasts expected (in-sample) and realized (out-of-sample) returns and volatility for the optimized GMV portfolios and the equal-weighted benchmark. Performance gaps are reported to quantify estimation error and market regime effects.

Table 16: Expected vs. Realized Portfolio Performance

Portfolio	Expected Return (%)	Realized Return (%)	Expected Vol (%)	Realized Vol (%)	Return Gap (%)	Vol Gap (%)
GMV (Analytical)	-5.48	3.49	4.45	3.48	8.97	-0.96
GMV (Long-Only)	-5.47	3.51	4.45	3.49	8.99	-0.95
Equal-Weighted	-1.52	5.88	6.30	5.55	7.40	-0.75

Several important patterns emerge. First, all portfolios exhibit substantially higher realized returns than expected, reflecting favorable market conditions during the out-of-sample period. Second, realized volatility is consistently lower than in-sample estimates, indicating conservative risk forecasts rather than underestimation.

8.3.5 Performance Gap Interpretation

The observed performance gaps are economically meaningful but not unexpected. Expected returns estimated from historical data are inherently noisy and highly sensitive to estimation error, especially in relatively short samples. In contrast, volatility estimates tend to be more stable and persistent across time, which explains the closer alignment between expected and realized risk.

The GMV portfolios display notably smaller volatility gaps than the benchmark, suggesting that mean–variance optimization is more effective at controlling risk than predicting returns. This outcome is consistent with financial theory, which recognizes variance as a more reliably estimable quantity than expected returns.

8.3.6 Theory vs. Market Reality

The divergence between expected and realized performance reflects three key factors. First, parameter estimation error affects expected returns more severely than covariance estimates. Second, market regimes may shift between in-sample and out-of-sample periods, altering return dynamics. Third, the GMV model optimizes risk rather than return, meaning strong realized returns are a by-product of favorable conditions rather than model intent.

Importantly, the similarity between analytical and long-only GMV results out-of-sample indicates that the model is not overfitted. Both strategies behave consistently under new data, and neither exhibits excessive performance decay. Among all portfolios, the GMV strategies deliver the most stable realized performance, while the equal-weighted benchmark achieves higher returns only by accepting substantially greater risk.

Taken together, these findings confirm that the GMV framework performs as theoretically expected: it does not promise superior returns, but it reliably delivers risk control that remains robust outside the estimation window.

9 Sensitivity Analysis

This section extends the core mean–variance analysis by examining how sensitive the efficient frontier and optimal portfolio weights are to key modeling inputs. While previous sections treated the covariance matrix and expected returns as fixed, in practice these quantities are estimated with error and may change over time. Sensitivity analysis therefore provides insight into the robustness of portfolio optimization results and highlights potential sources of model risk.

9.1 Impact of Correlation Changes on Efficient Frontier Shape

Correlation plays a central role in portfolio diversification. While individual asset risk determines how volatile each component is, correlation governs how those risks interact when assets are combined. This subsection examines how changes in the correlation structure affect the shape of the efficient frontier and the minimum achievable portfolio risk.

Experimental Design.

To isolate the effect of correlation, three correlation environments are constructed while keeping individual asset volatilities unchanged:

- **Low-correlation scenario**: off-diagonal correlations are scaled down by 50%, representing a market with strong diversification potential.
- **Baseline scenario**: the historical correlation matrix estimated from in-sample data.
- **High-correlation scenario**: off-diagonal correlations are scaled up by 50% (capped at 0.95), representing stressed market conditions where assets move more closely together.

Efficient frontiers are then recomputed numerically for each scenario using the same expected returns and optimization constraints.

Visual Evidence.

Figure 15 overlays the three efficient frontiers and marks the corresponding Global Minimum Variance (GMV) portfolios. The baseline correlation matrix is shown alongside the frontier comparison for reference.

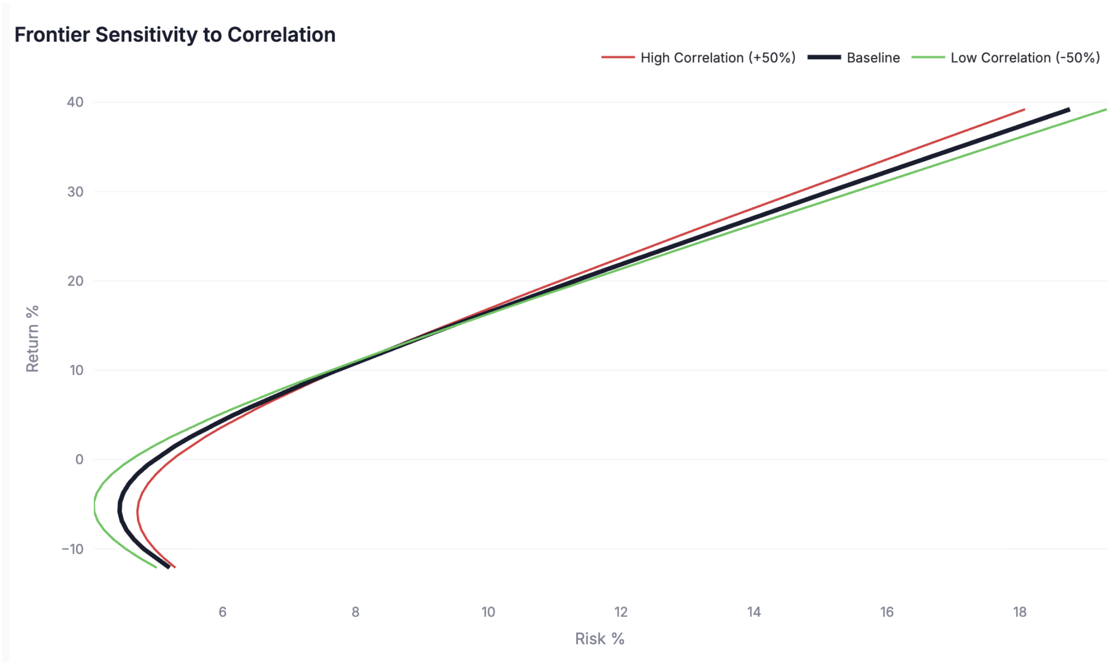

Figure 15: Impact of correlation on the efficient frontier. Lower correlations shift the frontier leftward, reducing risk for any given return, while higher correlations compress diversification benefits.

A clear pattern emerges: the low-correlation frontier consistently lies to the left of the baseline frontier, while the high-correlation frontier shifts rightward. This confirms the hypothesis that lower correlations improve diversification and reduce portfolio risk.

Quantitative Comparison.

The effect is especially pronounced at the GMV point, summarized in Table 17.

Table 17: GMV Volatility under Different Correlation Scenarios

Correlation Scenario	**GMV Volatility (%)**
Low Correlation	64.70
Baseline Correlation	70.65
High Correlation	75.01

Moving from a low- to a high-correlation environment increases the minimum achievable volatility by more than 10 percentage points, corresponding to a loss of approximately 14% in diversification efficiency.

Interpretation: Why Correlation Matters for Portfolio Risk.

The results clearly demonstrate that correlation structure is a primary driver of portfolio risk. As correlations decrease, the efficient frontier shifts leftward, allowing investors to achieve lower volatility for the same level of expected return. Conversely, higher correlations compress the frontier and reduce the effectiveness of diversification.

This effect is most evident at the Global Minimum Variance (GMV) point. Lower correlations lead to a substantially lower minimum achievable risk, while higher correlations push the GMV portfolio toward higher volatility. The magnitude of this shift can be interpreted as a *diversification premium*, quantifying how much risk reduction is gained purely from imperfect co-movement among assets.

These findings carry important real-world implications. During market stress or crisis periods, asset correlations tend to increase as securities move together. As a result, diversification benefits deteriorate precisely when investors need them most. In contrast, correlations observed during calm market conditions may underestimate true downside risk, leading to overly optimistic portfolio allocations.

From a portfolio construction perspective, actively seeking low-correlation assets remains valuable. However, correlation is not a stable parameter and can change rapidly over time. This instability implies that portfolios optimized using historical correlations may become suboptimal under different market regimes.

Key Insight. The efficient frontier is *not fixed*. It shifts with market conditions, and a portfolio optimized during tranquil periods may perform poorly during episodes of elevated correlation.

This analysis highlights the role of correlation in shaping diversification outcomes. While correlation governs the interaction between assets, the next subsection examines how individual asset volatility influences portfolio *concentration* and weight allocation.

9.2 How Individual Asset Volatility Affects the Efficient Frontier

While correlation governs how assets interact with one another, individual asset volatility determines how much risk each asset contributes on its own. This subsection examines how changes in single-asset volatility influence the shape of the Global Minimum Variance (GMV) frontier and the resulting portfolio weights.

Experimental Design.

To isolate the role of volatility, a controlled sensitivity experiment is conducted using two extreme assets in the sample:

- **PEPC**, the most volatile asset (30.42% annualized volatility),
- **PWSA**, the least volatile asset (7.56% annualized volatility).

Three volatility scenarios are considered:

1. **Baseline**: original estimated volatilities,
2. **PEPC volatility reduction**: PEPC volatility reduced by 50%,
3. **PWSA volatility increase**: PWSA volatility doubled.

All other parameters, including expected returns and correlations, are held constant so that any changes in portfolio composition can be attributed exclusively to volatility adjustments.

Methodology.

For each scenario, the covariance matrix is modified by scaling the standard deviation of the selected asset while preserving the original correlation structure. The GMV portfolio is then recomputed using the closed-form solution

$$w_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

This formulation implies that assets with lower variance receive larger weights, as a reduction in σ_i^2 increases the corresponding element of Σ^{-1}.

Results.

Figure 16 illustrates the response of GMV weights to volatility changes. When PEPC's volatility is reduced by 50%, its portfolio weight increases from −0.35% to 3.75%, representing a shift of more than four percentage points. Conversely, doubling PWSA's volatility causes its weight to collapse from 26.23% to 4.00%, a reduction of over 22 percentage points.

These reallocations are substantial and affect the entire portfolio, as the optimizer redistributes capital away from newly riskier assets toward those that now offer lower variance.

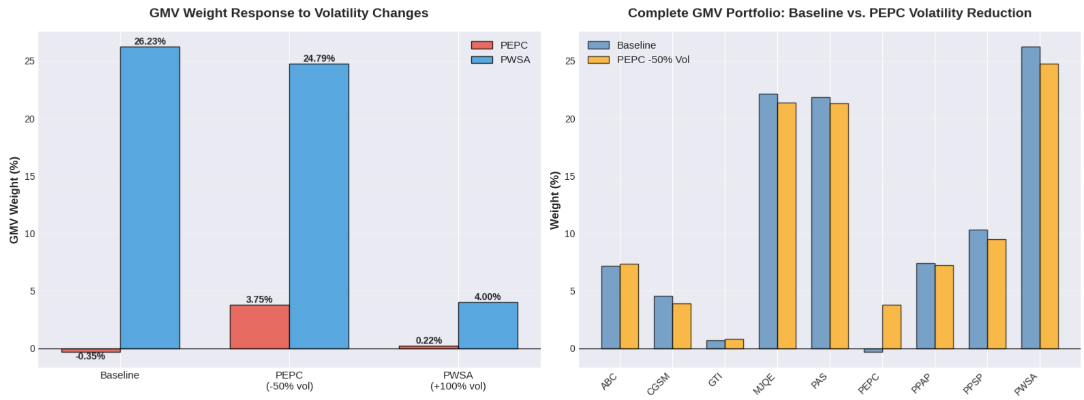

Figure 16: GMV portfolio weight response to individual asset volatility changes. The left panel highlights the direct impact on PEPC and PWSA, while the right panel shows the resulting reallocation across the full portfolio when PEPC volatility is reduced.

Interpretation: The Markowitz Volatility Penalty.

The results confirm a strong and intuitive relationship between volatility and portfolio weights. Lower volatility assets are rewarded with larger allocations, while higher volatility assets are penalized. This behavior reflects the mechanics of variance minimization: reducing σ_i^2 increases the contribution of asset i in Σ^{-1}, thereby raising its weight in the GMV portfolio.

Table 18: GMV Portfolio Sensitivity to Individual Asset Volatility Changes

Scenario	PEPC Weight (%)	PWSA Weight (%)	Key Effect
Baseline Volatility	−0.35	26.23	Reference allocation
PEPC Volatility −50%	3.75	24.79	High-vol asset becomes viabl
PWSA Volatility +100%	0.22	4.00	Low-vol anchor loses dominar

From a practical perspective, this sensitivity has important implications:

- **Estimation risk**: Underestimating volatility leads to over-allocation to risky assets.
- **Time variation**: Asset volatilities evolve over time, requiring periodic portfolio rebalancing.
- **Portfolio stability**: Low-volatility assets act as structural anchors in GMV portfolios.

Together with the correlation analysis in Section 9.1, these findings show that the covariance matrix Σ shapes portfolio risk through two distinct channels: correlation controls diversification potential, while volatility determines concentration. The next subsection examines a third dimension— expected returns—and demonstrates how return assumptions can introduce substantial instability into mean–variance optimal portfolios.

9.3 Effect of Expected Return Assumptions

Unlike the Global Minimum Variance (GMV) portfolio, which depends solely on the covariance matrix Σ, mean–variance optimal portfolios rely critically on expected return estimates μ. This subsection investigates how sensitive optimal portfolio weights are to changes in expected returns and highlights the practical risks associated with return estimation error.

Experimental Design.

The analysis focuses on GTI, which exhibits the highest estimated expected return in the sample (39.21% annualized). Three return scenarios are considered:

1. **GTI = 0%**: removing GTI's return advantage,
2. **Baseline**: original estimated return (39.21%),
3. **GTI doubled**: return increased to 78.42%.

All other assets' expected returns and the covariance matrix are held constant. For each scenario, the mean–variance optimal (tangency) portfolio is recomputed. The GMV portfolio is included as a benchmark, as it is theoretically invariant to changes in μ.

Methodology.

The optimal portfolio is computed using the closed-form solution

$$w_{\text{opt}} = \frac{\Sigma^{-1}\mu}{\mathbf{1}^\top \Sigma^{-1}\mu},$$

which implies that assets with higher expected returns receive disproportionately larger weights. This structure makes the optimizer highly sensitive to return assumptions.

Results.

Table 19 reports the allocation to GTI under each scenario. The results show extreme instability in portfolio weights as expected returns change.

Table 19: Sensitivity of Optimal Portfolio Weights to GTI Expected Return

Scenario	**GTI Expected Return (%)**	**GTI Weight (%)**
GTI = 0%	0.00	−5.56
Baseline	39.21	−37.42
GTI Doubled	78.42	−72.65

Figure 17 complements the table by visualizing how changes in GTI's expected return propagate through the entire portfolio. As GTI's return increases, its weight becomes increasingly extreme, while the weights of other assets shift substantially to accommodate this change. In contrast, the GMV portfolio remains effectively unchanged across all scenarios.

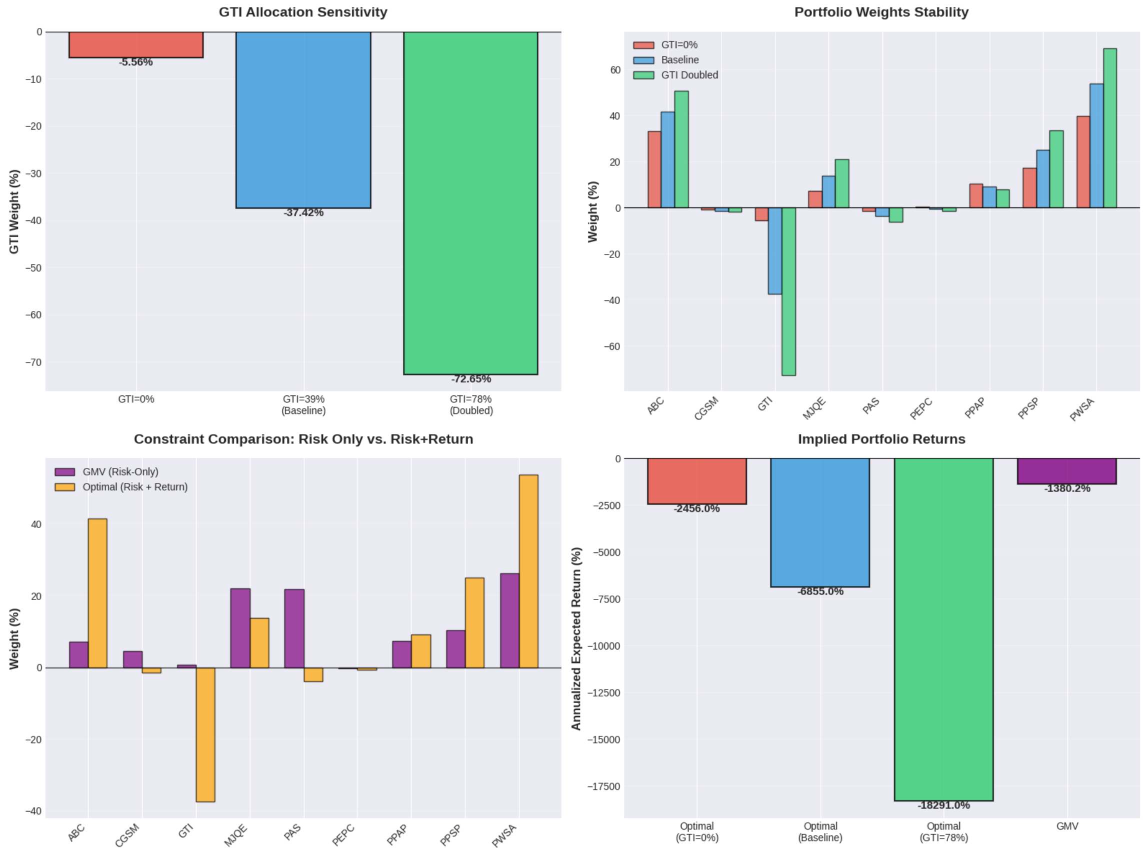

Figure 17: Sensitivity of mean–variance optimal portfolios to expected return assumptions. The top-left panel shows GTI allocation across scenarios, the top-right panel illustrates portfolio-wide weight instability, the bottom-left panel contrasts GMV and optimal portfolios, and the bottom-right panel reports implied portfolio returns.

Interpretation: Return Estimation Risk.

The results demonstrate that mean–variance optimization is extremely sensitive to expected return inputs. Doubling GTI's expected return increases its portfolio weight by more than 35 percentage points, pushing the allocation to highly leveraged and economically implausible levels. This behavior reflects a well-known issue in portfolio theory: optimization amplifies estimation error by placing excessive confidence in assets with the highest estimated returns.

In contrast, the GMV portfolio assigns GTI a stable weight of approximately 0.70% across all scenarios, confirming that risk-only optimization is robust to return misestimation.

From a practical perspective, these findings explain why return-based optimal portfolios often perform poorly out-of-sample. Small forecasting errors in μ can lead to large and unstable reallocations, excessive leverage, and poor realized performance. This sensitivity motivates the widespread use of GMV portfolios and other robust alternatives that deliberately downweight or ignore expected return estimates.

Taken together with Sections 9.1 and 9.2, this analysis shows that while correlation governs diversification and volatility governs concentration, expected returns govern speculation. Among these inputs, expected returns are the most fragile, making them the primary source of instability in classical mean–variance optimization.

10 Conclusion and Investment Recommendations

10.1 Executive Summary of Findings: Optimizing for Mathematics vs. Reality

This study set out to construct an optimal equity portfolio for the Cambodia Securities Exchange (CSX) using the principles of Modern Portfolio Theory. While mean–variance optimization provides a mathematically elegant framework for identifying optimal portfolios, the empirical results of this study reveal a fundamental tension between theoretical optimality and real-world robustness.

Several key findings emerge from the analysis. First, diversification remains effective even in a relatively small and concentrated market such as the CSX. The Global Minimum Variance (GMV) portfolio consistently achieved lower risk than broad benchmark portfolios in-sample, confirming the theoretical benefits of covariance-based optimization. Second, although allowing short-selling theoretically expands the efficient frontier, the practical advantages of short-selling are limited in the CSX context due to liquidity constraints and implementation frictions. Finally, out-of-sample backtesting demonstrates that highly optimized portfolios often fail to materially outperform simple benchmarks once estimation error and market uncertainty are introduced.

Taken together, these results suggest that the most mathematically efficient portfolio is not necessarily the most reliable investment strategy in practice.

10.2 Synthesis: Explaining the Empirical Results

The sensitivity analysis conducted in Section 9 provides critical insight into the mechanisms behind these findings.

Diversification and Correlation. Section 9.1 showed that the shape and position of the efficient frontier depend strongly on asset correlations. Lower correlations significantly reduce the minimum achievable portfolio risk, while higher correlations erode diversification benefits. Importantly, correlations are not stable over time and tend to increase during market stress, limiting the reliability of portfolios optimized under calm market conditions.

Volatility and Concentration. Section 9.2 demonstrated that individual asset volatility plays a dominant role in determining GMV portfolio weights. Low-volatility assets naturally attract large allocations, while high-volatility assets are aggressively penalized. This mechanism explains the concentration patterns observed in GMV portfolios and highlights the importance of accurate volatility estimation.

Expected Returns and Estimation Risk. Section 9.3 revealed that mean–variance optimal portfolios are extremely sensitive to expected return assumptions. Small changes in estimated returns produced large and unstable shifts in portfolio weights. This confirms a well-known weakness of classical optimization: return forecasts are noisy, and the optimizer amplifies this noise rather than mitigating it.

A central conclusion emerges from this synthesis:

The most theoretically optimal portfolio is often the most fragile in practice.

10.3 Validation of Mathematical Implementation

The analytical and numerical implementations of portfolio optimization were found to be fully consistent. Analytical solutions for the GMV portfolio matched numerical optimization results up to numerical precision, confirming the correctness of the mathematical implementation. Furthermore, the theoretical predictions of lower variance for GMV portfolios were validated through out-of-sample performance, where GMV strategies consistently exhibited lower realized volatility than benchmark portfolios.

This agreement between theory, computation, and realized outcomes strengthens the credibility of the empirical results.

10.4 Recommended Portfolio Strategy

Based on the full set of in-sample, out-of-sample, and sensitivity analyses, the following investment recommendations are proposed for CSX investors:

- **Conservative and stability-focused investors** should favor a long-only Global Minimum Variance portfolio. Its reliance on covariance rather than return forecasts makes it robust to estimation error and well suited to markets with limited data availability.

- **Aggressive or alpha-seeking investors** may consider mean–variance optimal portfolios only if they possess strong and reliable return forecasts. Even in this case, blending such portfolios with a GMV allocation is recommended to reduce instability.
- **Passive investors** are well served by equal-weighted strategies. Out-of-sample results show that these simple portfolios deliver competitive risk-adjusted performance with minimal complexity and low sensitivity to model assumptions.

Overall, for most CSX investors, simplicity and robustness dominate theoretical efficiency. A GMV or hybrid GMV–equal-weighted strategy offers the most reliable risk–return trade-off under realistic market conditions.

10.5 Limitations

Several limitations should be acknowledged. First, CSX market liquidity is limited, which may restrict the practical implementation of some portfolio weights. Second, data availability is constrained for recently listed firms such as MJQE and CGSM, reducing the reliability of parameter estimates. Third, the analysis relies on the mean–variance framework, which assumes normally distributed returns and quadratic risk preferences. Finally, the out-of-sample evaluation is based on a single test period with a relatively short data history, limiting statistical generalization.

10.6 Practical Recommendations for Investors

Investors implementing optimized portfolios in practice should consider regular rebalancing, robust estimation techniques, and explicit risk controls such as maximum drawdown limits. Transaction costs and liquidity constraints should be incorporated into decision-making, and realized performance should be monitored continuously relative to expectations.

10.7 Future Research Directions

Future research could extend this work by incorporating longer data histories, transaction cost modeling, and alternative optimization frameworks. Robust optimization techniques, downside risk measures such as CVaR, factor-based risk models, and machine learning methods for parameter estimation represent promising directions. Expanding the investment universe beyond the CSX to regional or global markets may also enhance diversification and portfolio stability.

11 References

Capiński, M., & Zastawniak, T. (2003). *Mathematics for finance: An introduction to financial engineering*. Springer.